Phase 1 - Week 5 Side Talk

Hard things under your Cloud

Audience: XBrain Class of 2026

By: Nam Hong
May 14th, 2026

Topics

- Hard things enabling VPC

Another Day, Another Billion Packets

Eric Brandwine

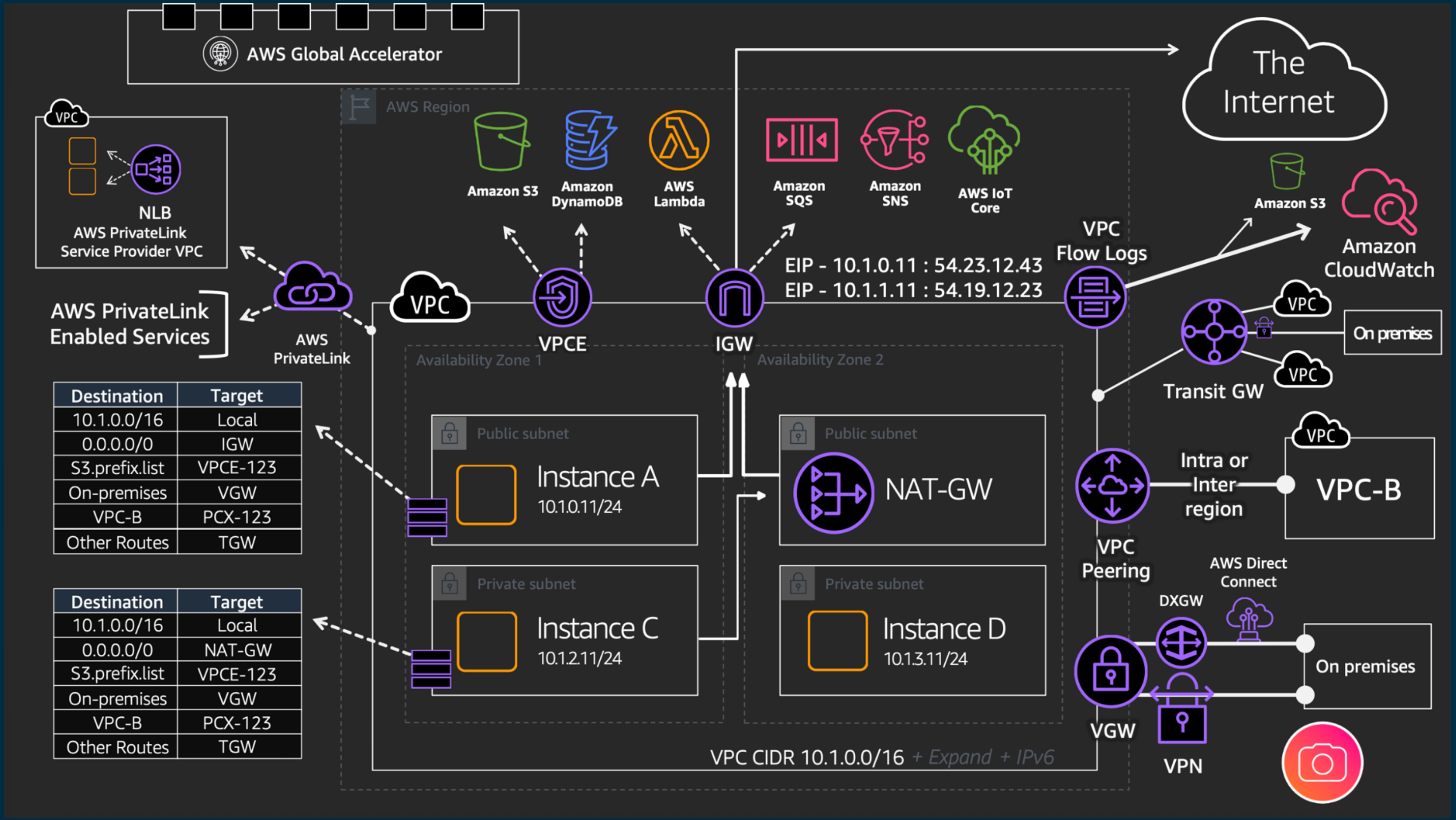
https://www.youtube.com/watch?v=3qln2u1Vr2E

Another Day, Another Billion Flows

Colm MacCarthaigh

https://www.youtube.com/watch?v=8gc2DgBqo9U

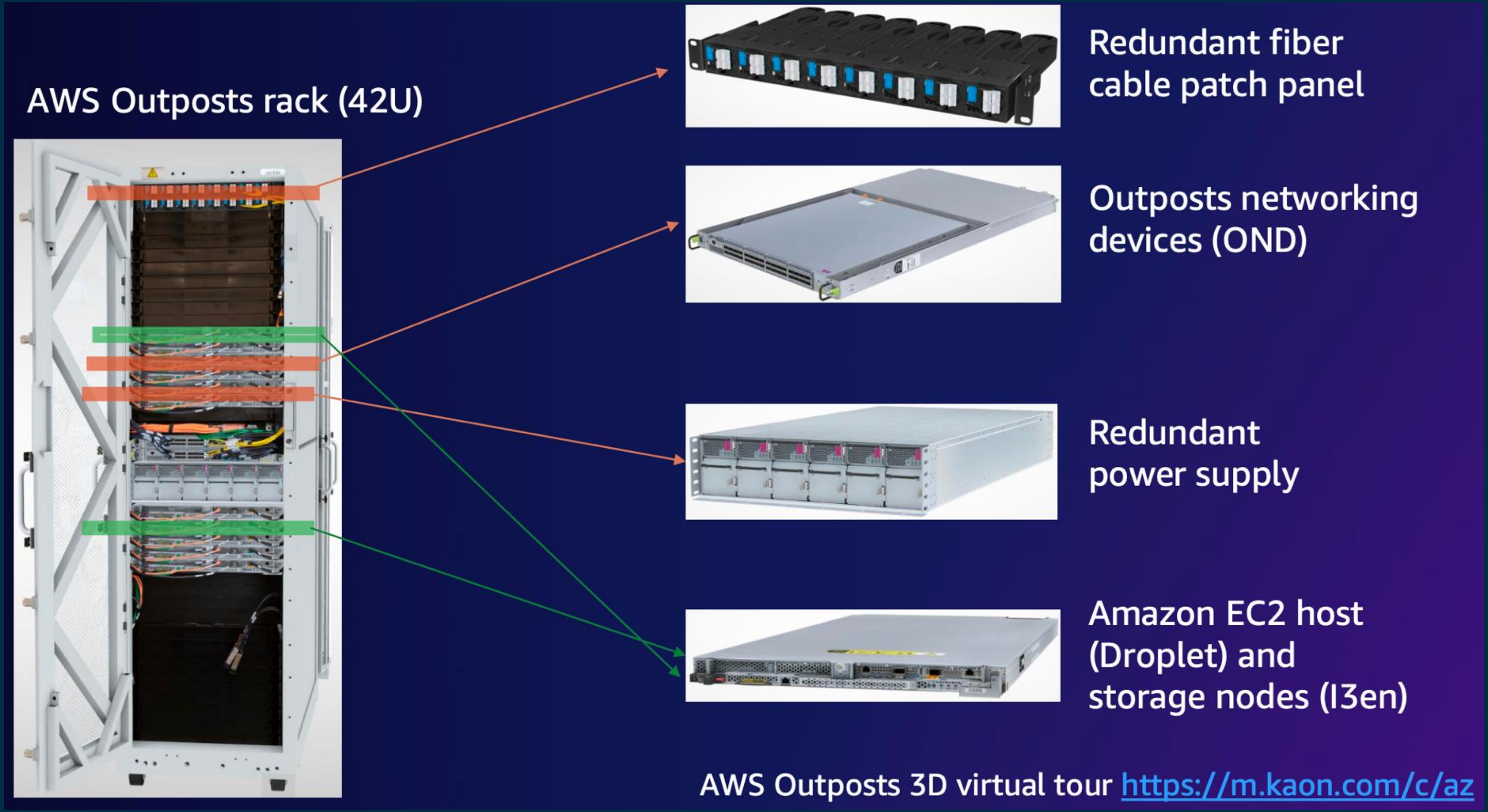
VPC Networking



Destination	Target
10.1.0.0/16	Local
0.0.0.0/0	IGW
S3.prefix.list	VPCE-123
On-premises	VGW
VPC-B	PCX-123
Other Routes	TGW

Destination	Target
10.1.0.0/16	Local
0.0.0.0/0	NAT-GW
S3.prefix.list	VPCE-123
On-premises	VGW
VPC-B	PCX-123
Other Routes	TGW

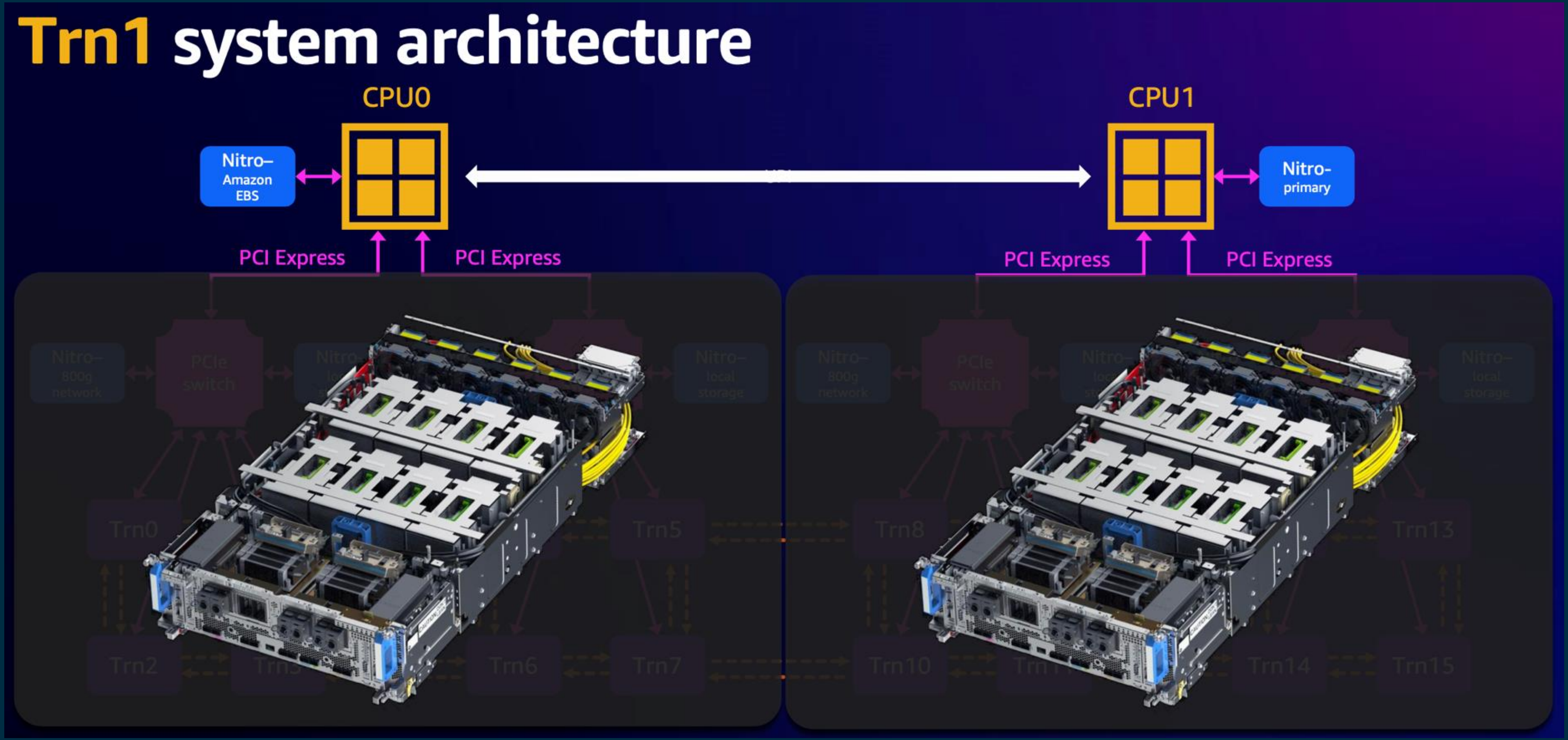
VPC Networking



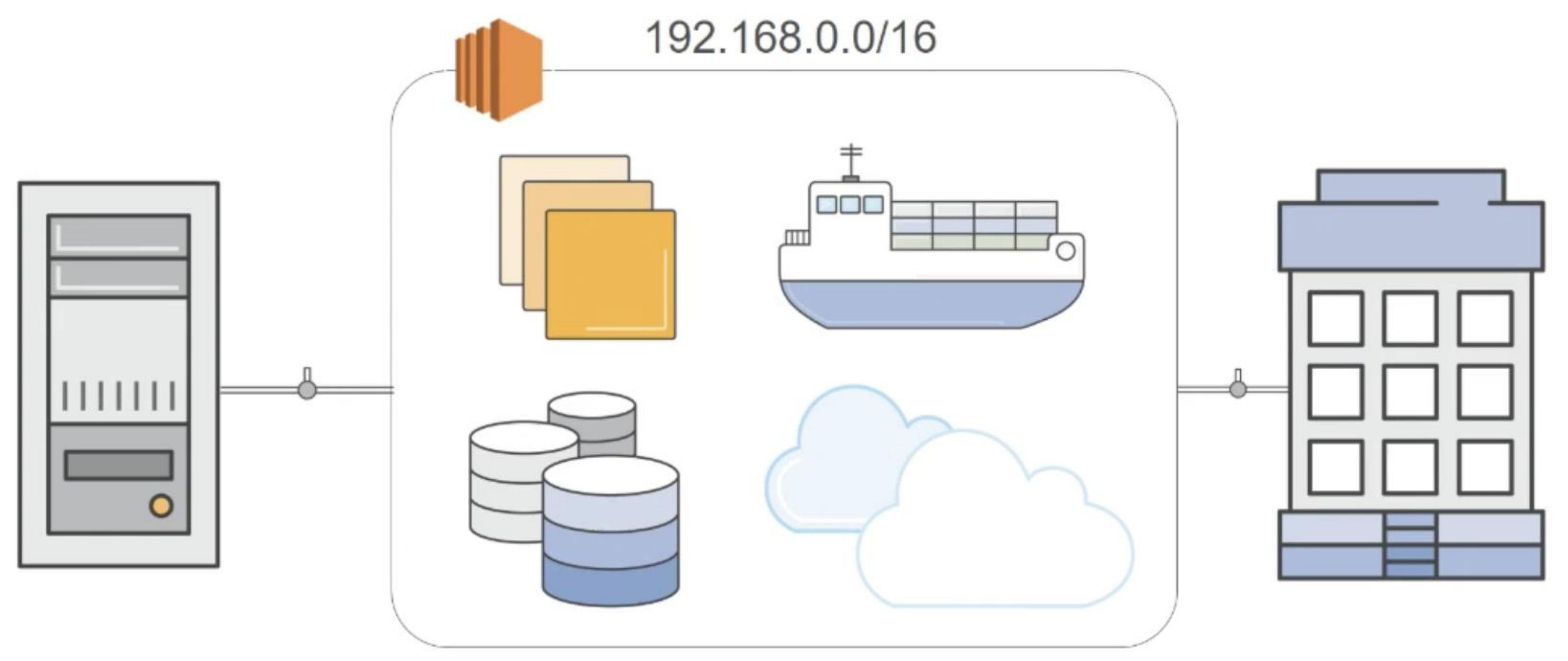
AWS Nitro

Nitro System
5 generations of AWS-built Nitro Chips and Cards
NITRO 1
NITRO 2
NITRO 3
NITRO 4
NITRO 5

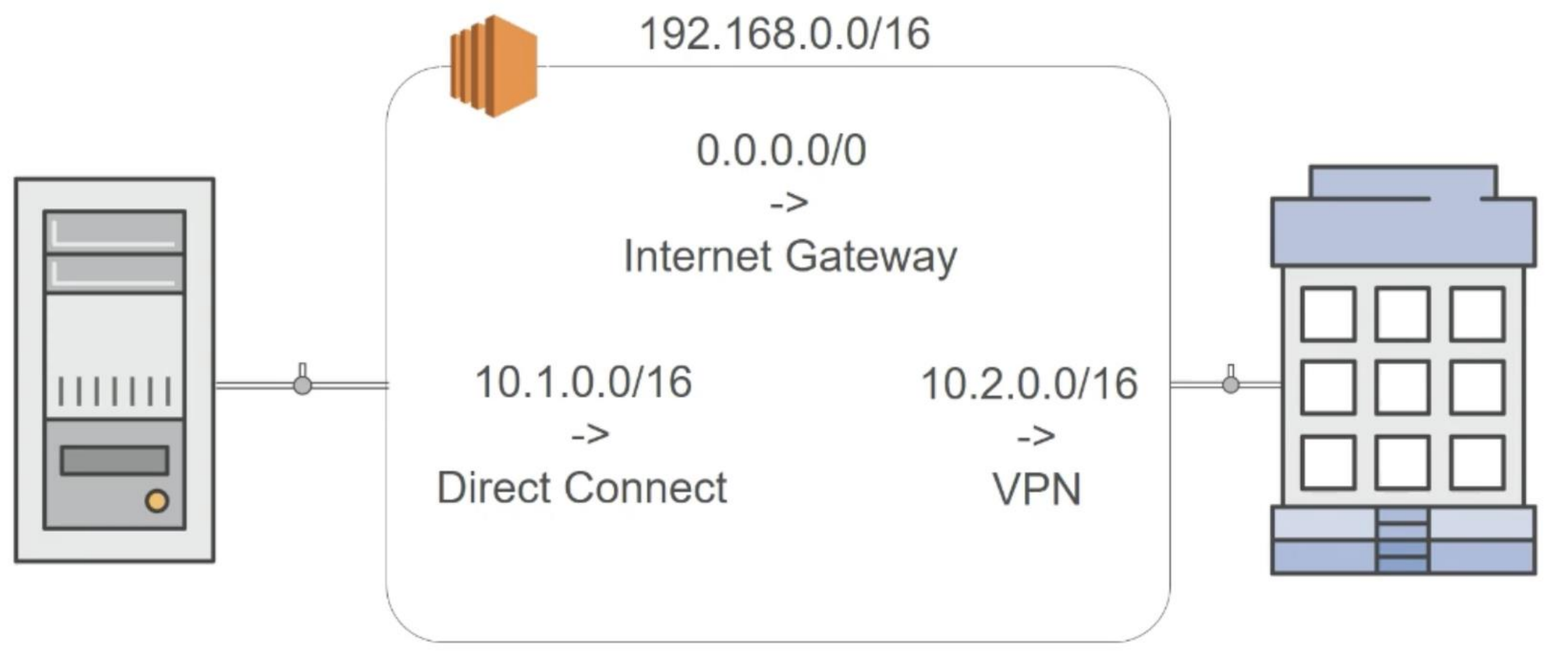
AWS Trainium



What is VPC?



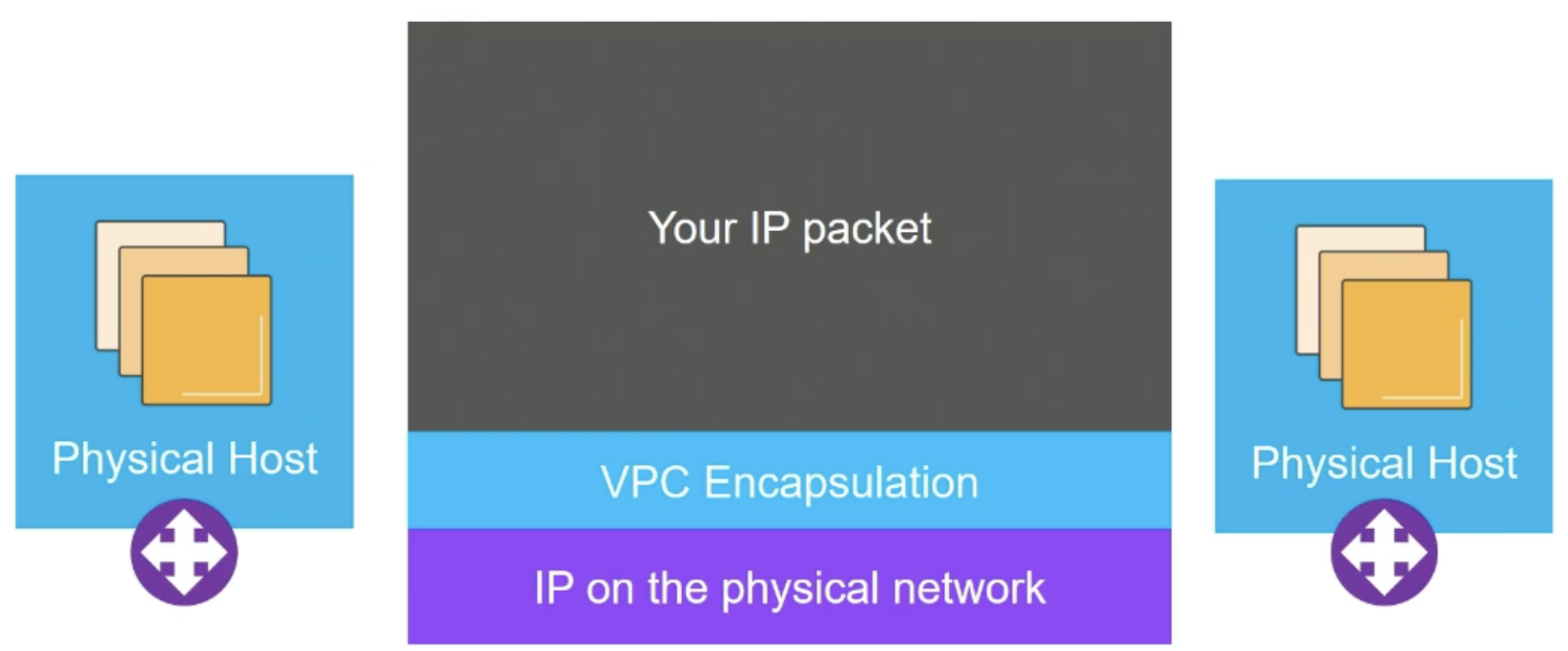
What is VPC?


192.168.0.0/16
0.0.0.0/0
->
Internet Gateway
10.1.0.0/16
->
Direct Connect
10.2.0.0/16
->
VPN

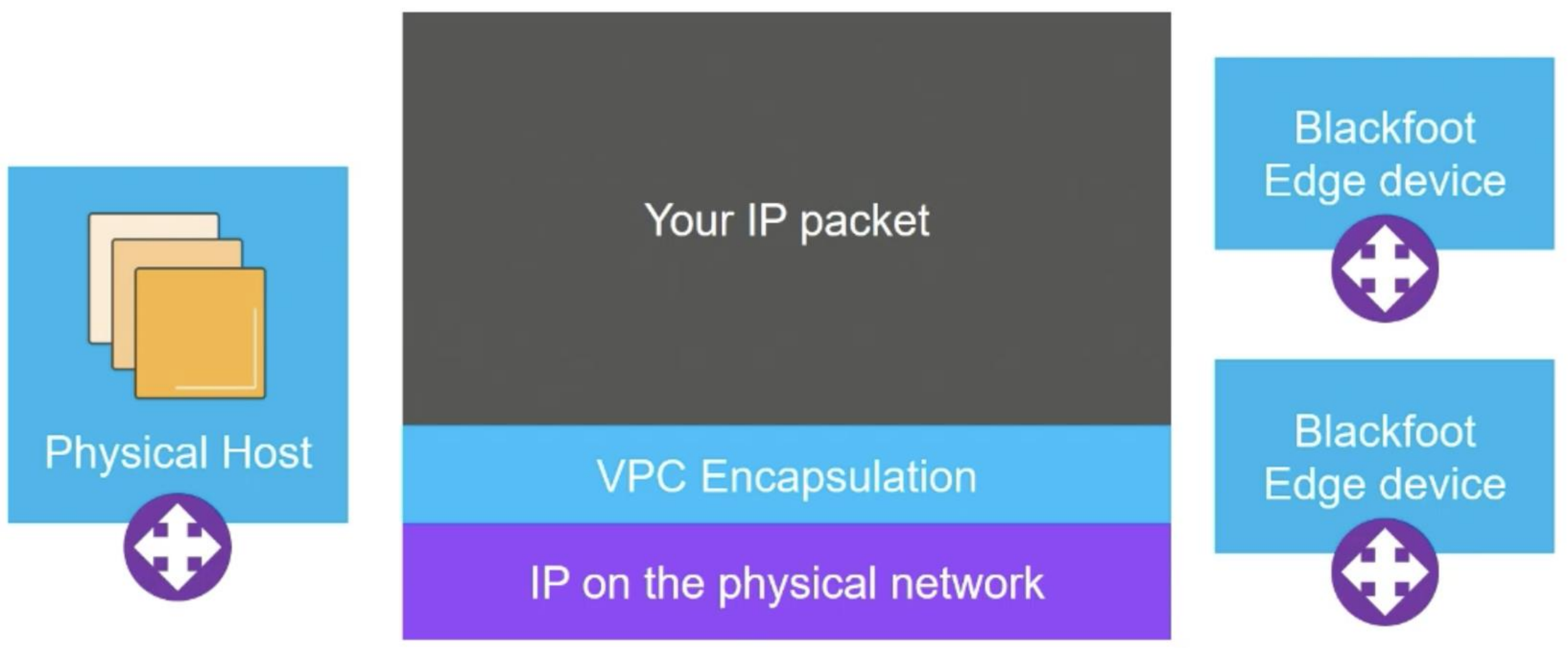
Every VPC comes with..

- Full programmatic control via APIs, templates, change history and audit capabilities, flow log support
- Built-in DHCP and DNS service, including private DNS
- Built-in firewall
- 9001 byte MTU

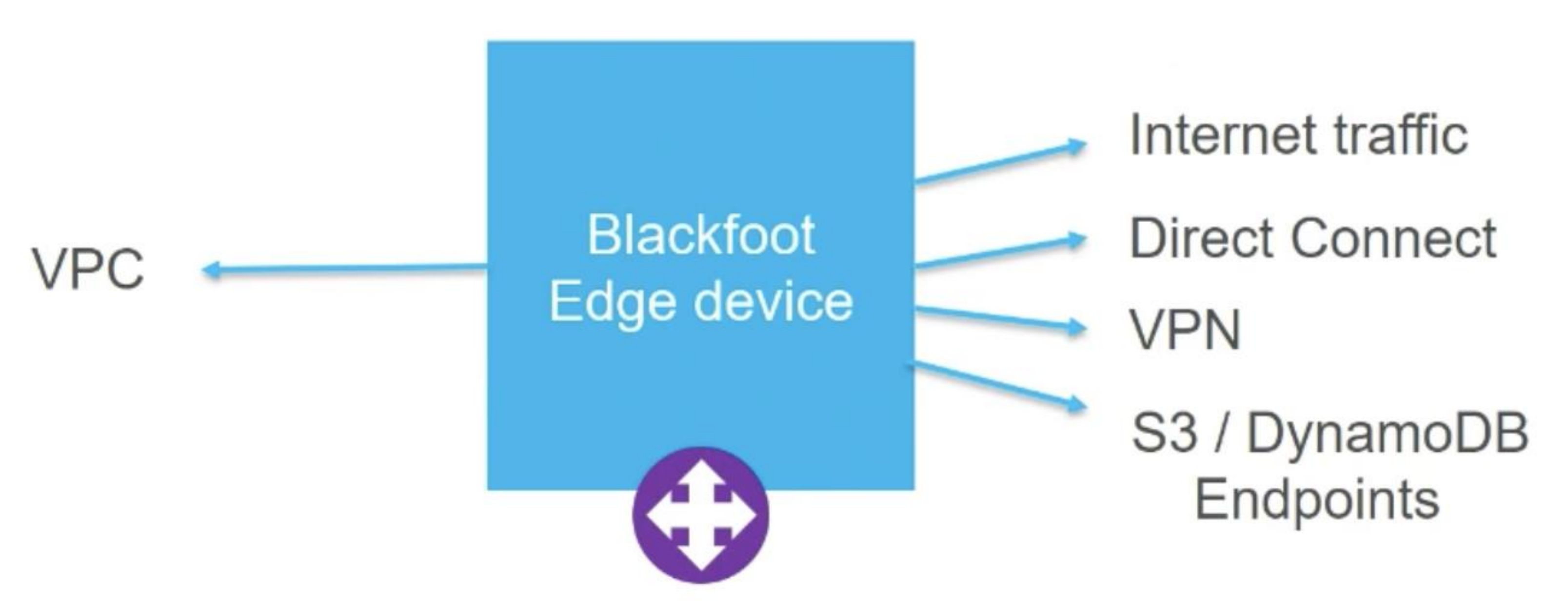
VPC on the wire



VPC on the wire



VPC on the wire



Encapsulating the packet

Outer-most IP destination identifies the target physical host

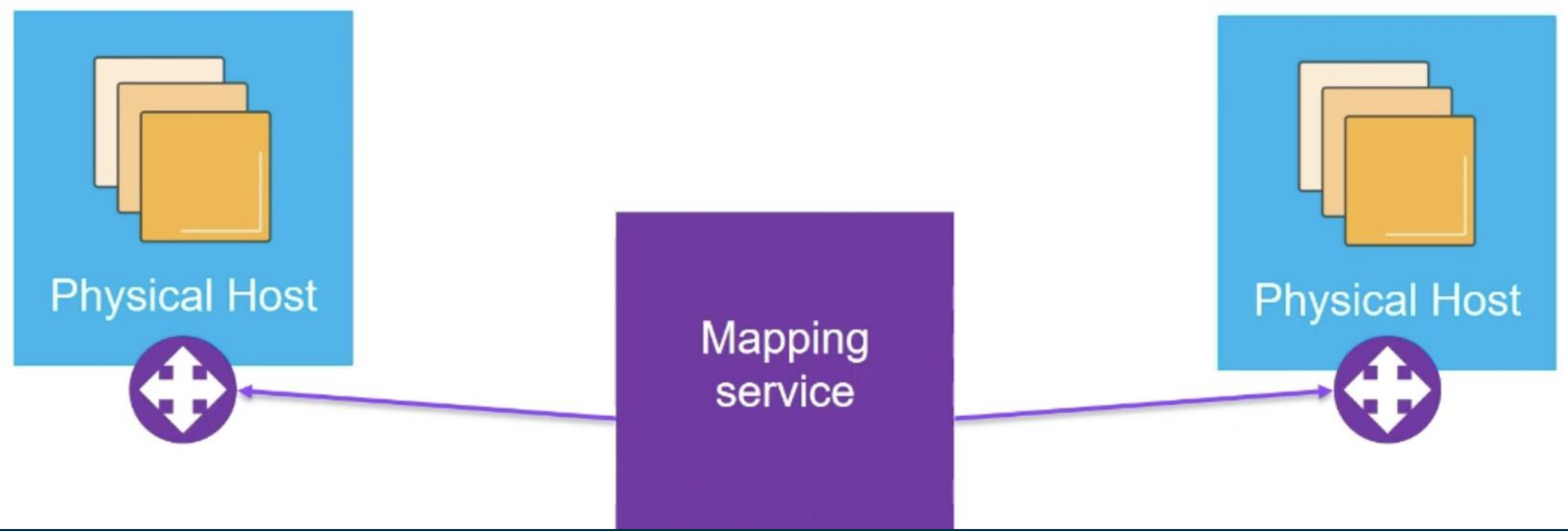
Encapsulation marks each packet with the VPC and the Elastic Network Interface

How does the sender know these? The mapping service

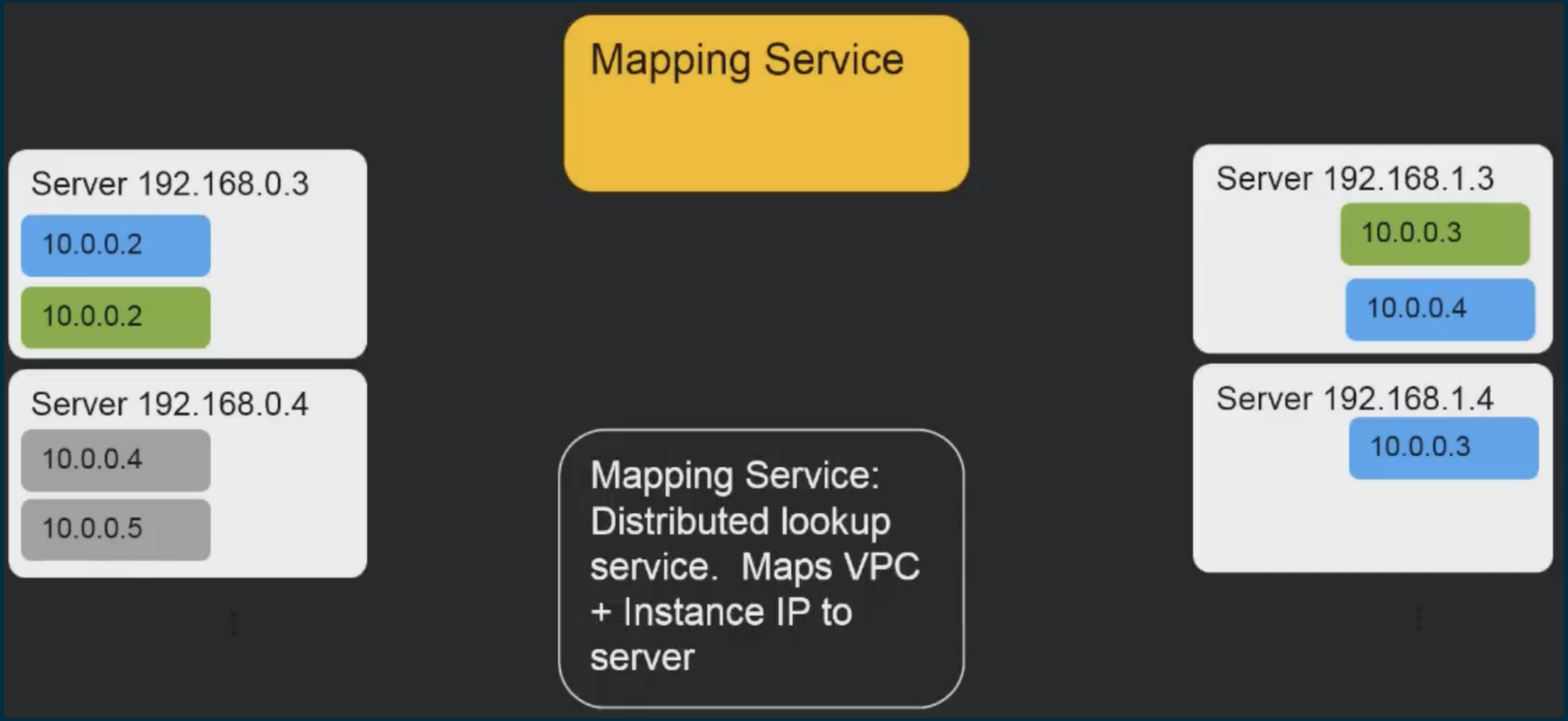
The mapping service

- A distributed web service that handles mappings between customers VPC routes and IPs and physical destinations on the wire.
- To support microsecond-scale latencies, mappings are cached where they are used, and pro-actively invalidated when they change.

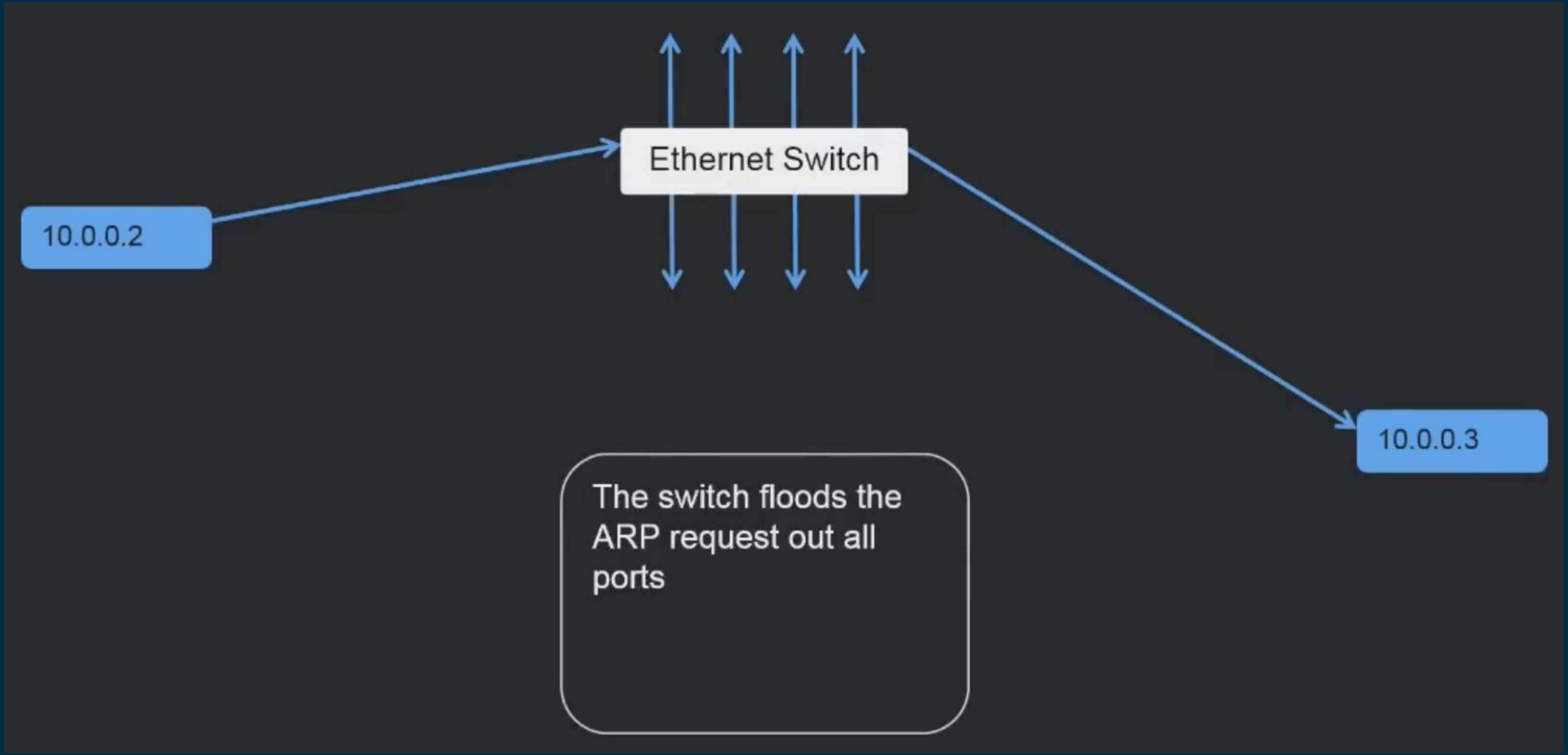
The mapping service



The mapping service

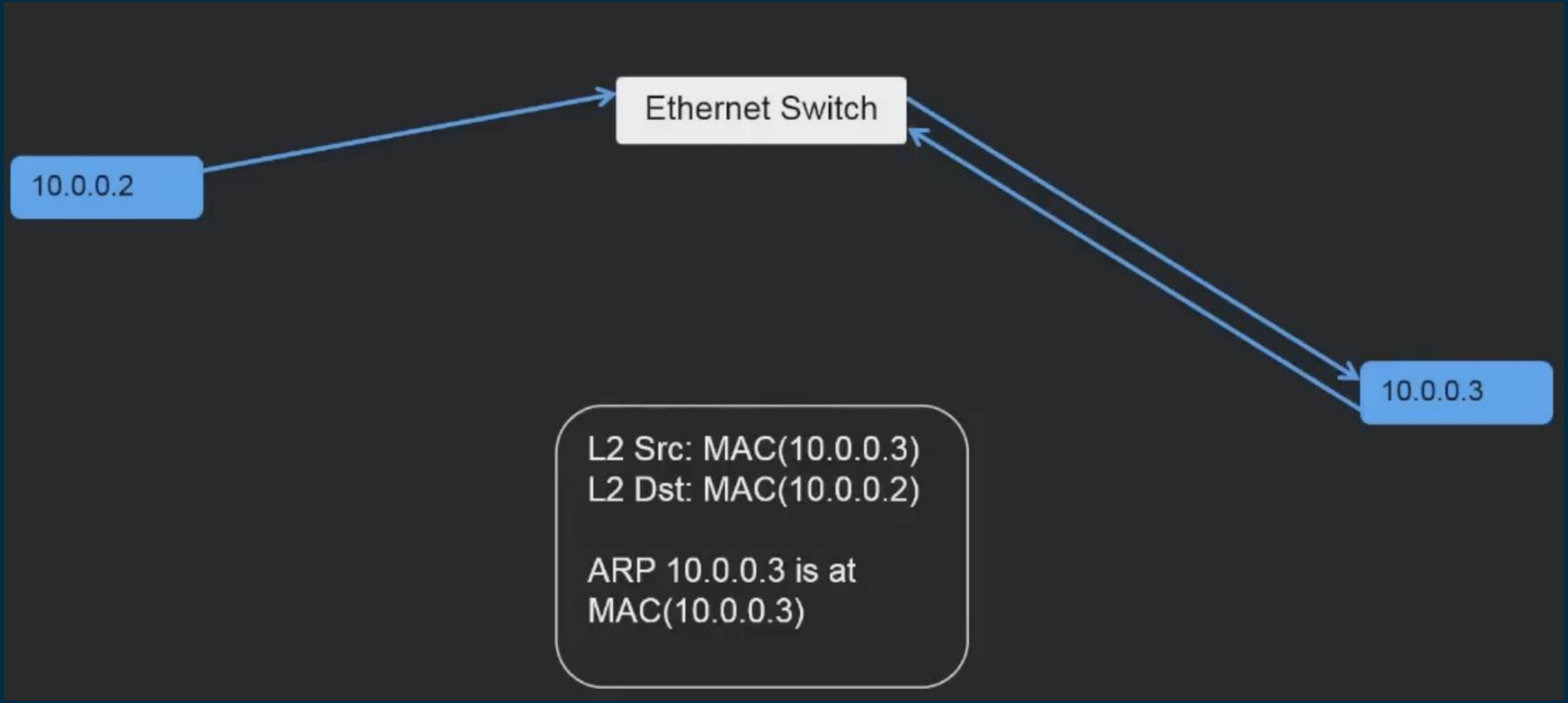


L2 - Ethernet



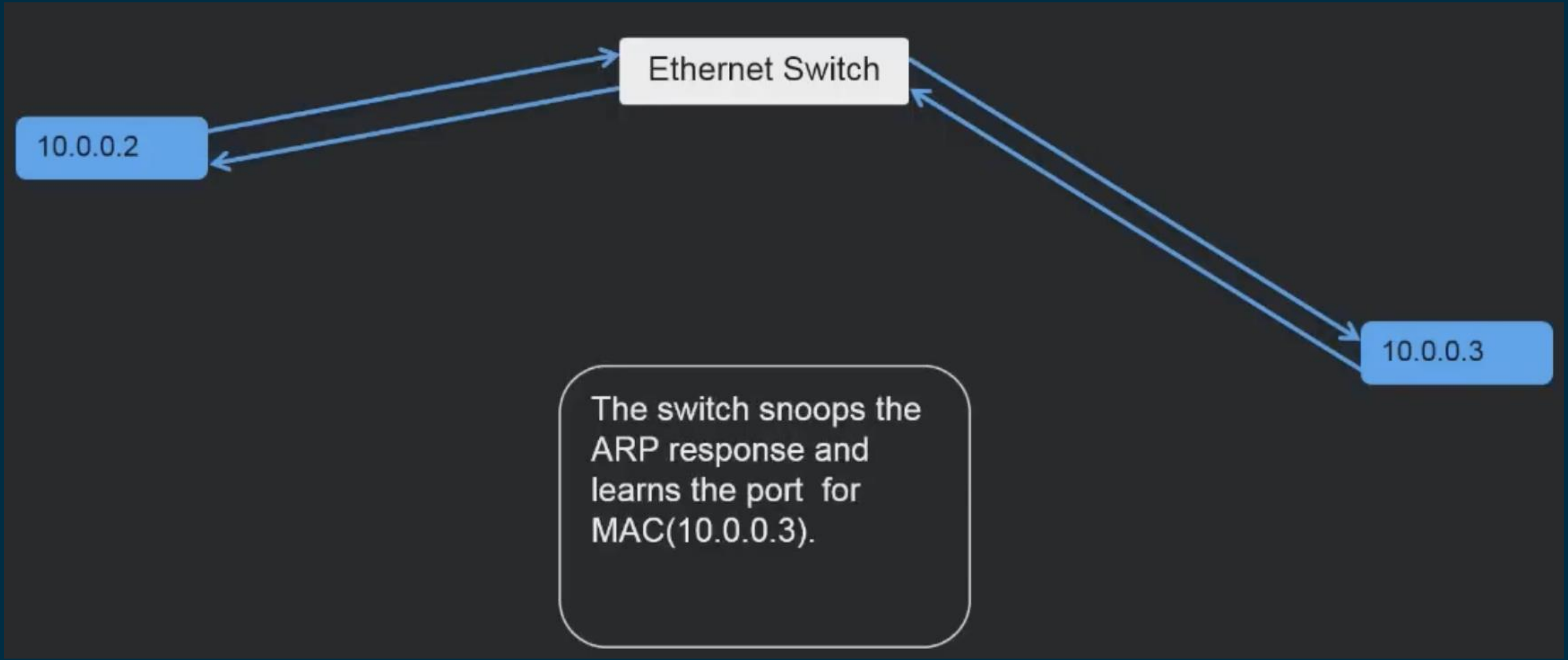
Ethernet Switch
10.0.0.2
10.0.0.3
The switch floods the ARP request out all ports

L2 - Ethernet

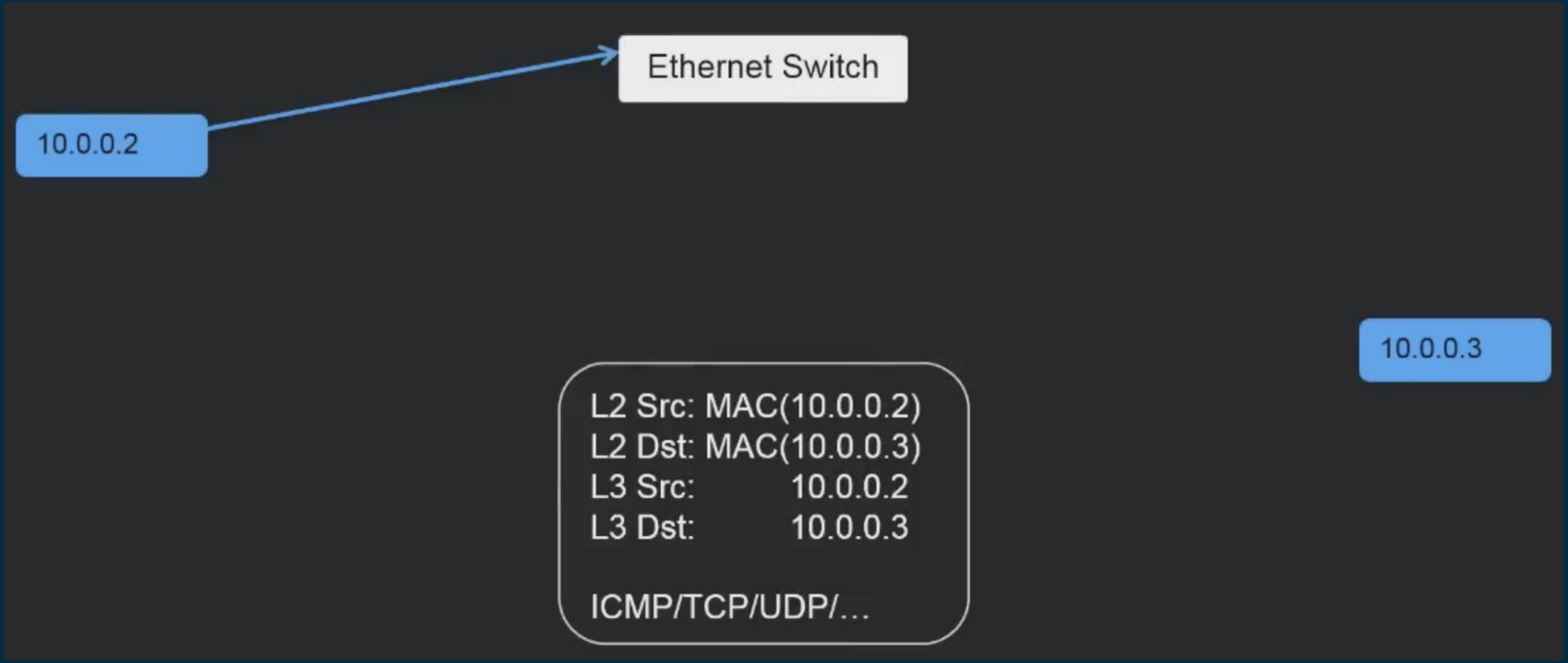


Ethernet Switch
10.0.0.2
10.0.0.3
L2 Src: MAC(10.0.0.3)
L2 Dst: MAC(10.0.0.2)
ARP 10.0.0.3 is at
MAC(10.0.0.3)

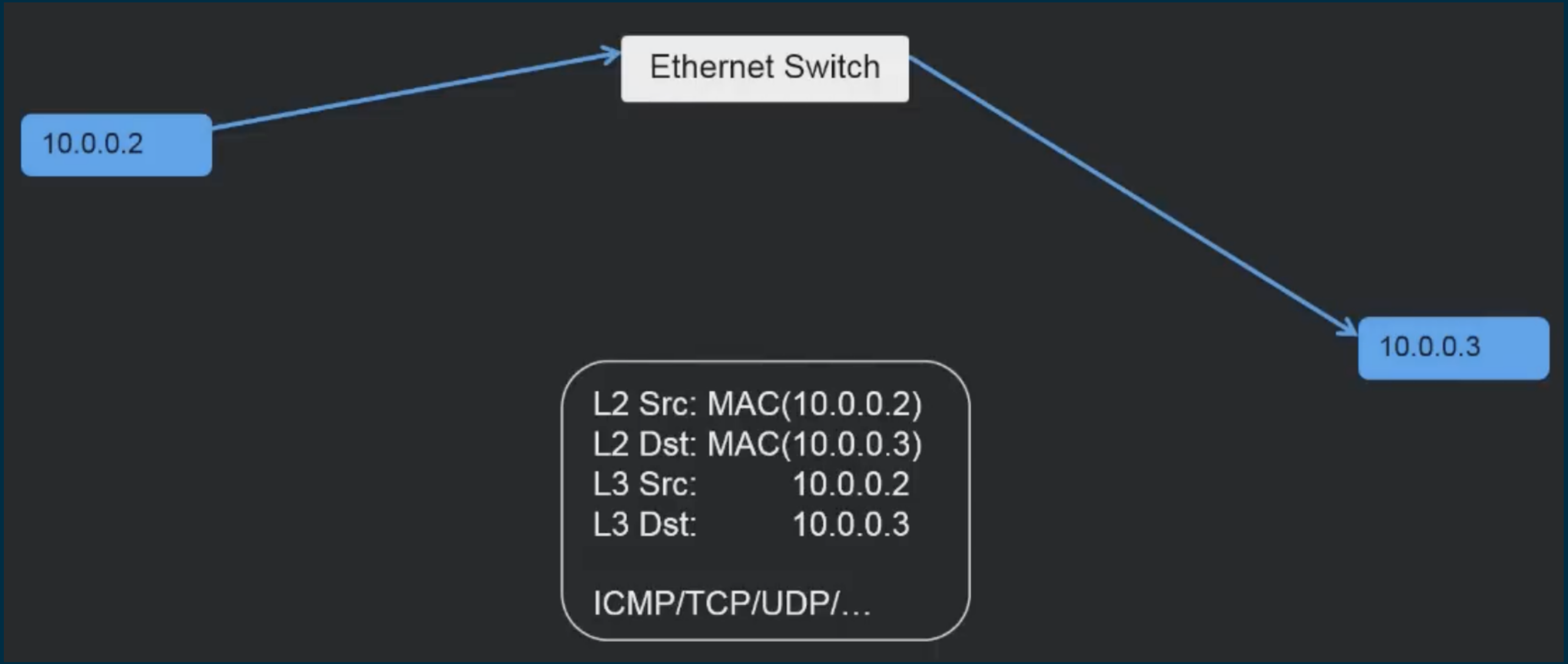
L2 - Ethernet


Ethernet Switch
10.0.0.2
10.0.0.3
The switch snoops the ARP response and learns the port for MAC(10.0.0.3).

L2 - Ethernet

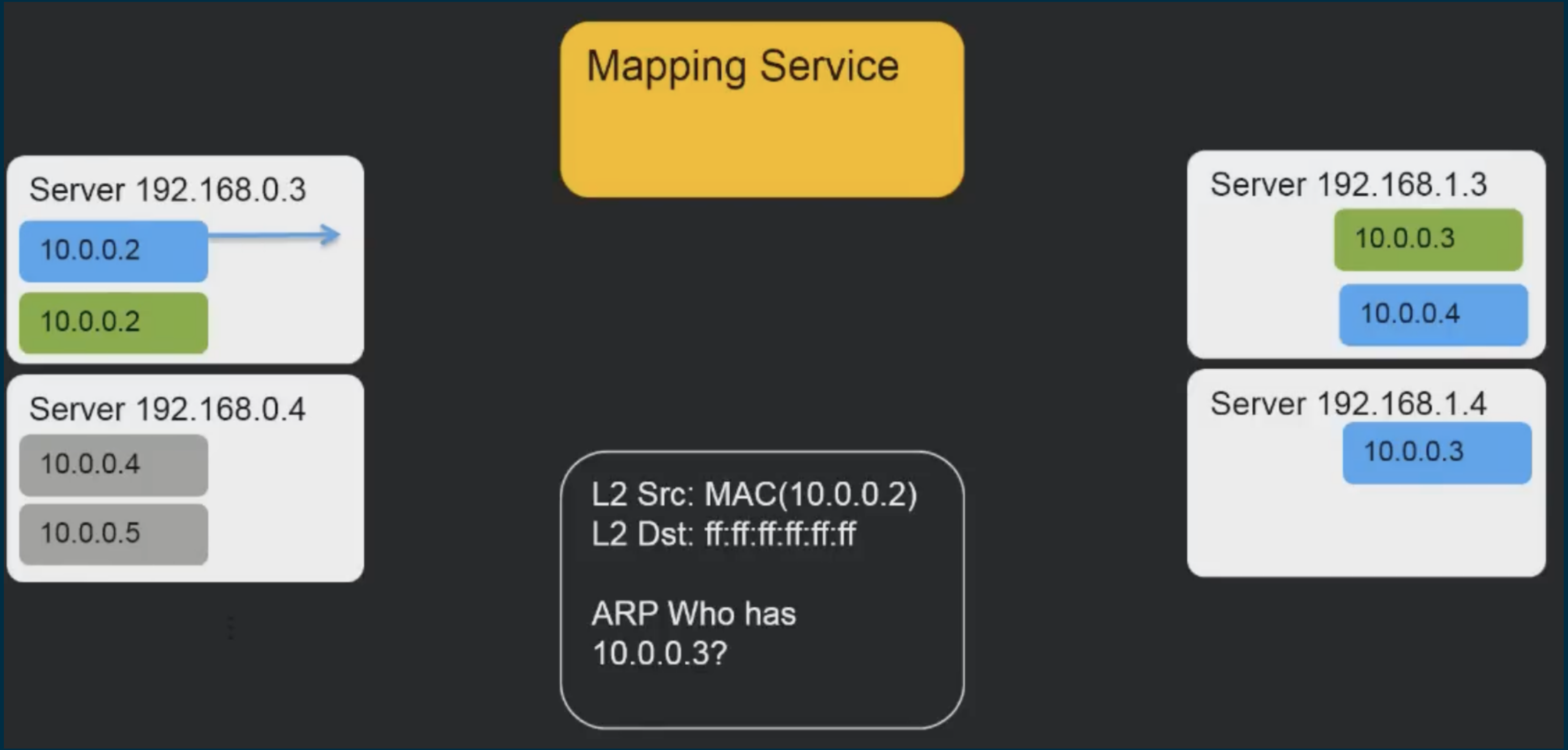

Ethernet Switch
10.0.0.2
10.0.0.3
L2 Src: MAC(10.0.0.2)
L2 Dst: MAC(10.0.0.3)
L3 Src: 10.0.0.2
L3 Dst: 10.0.0.3
ICMP/TCP/UDP/...

L2 - Ethernet

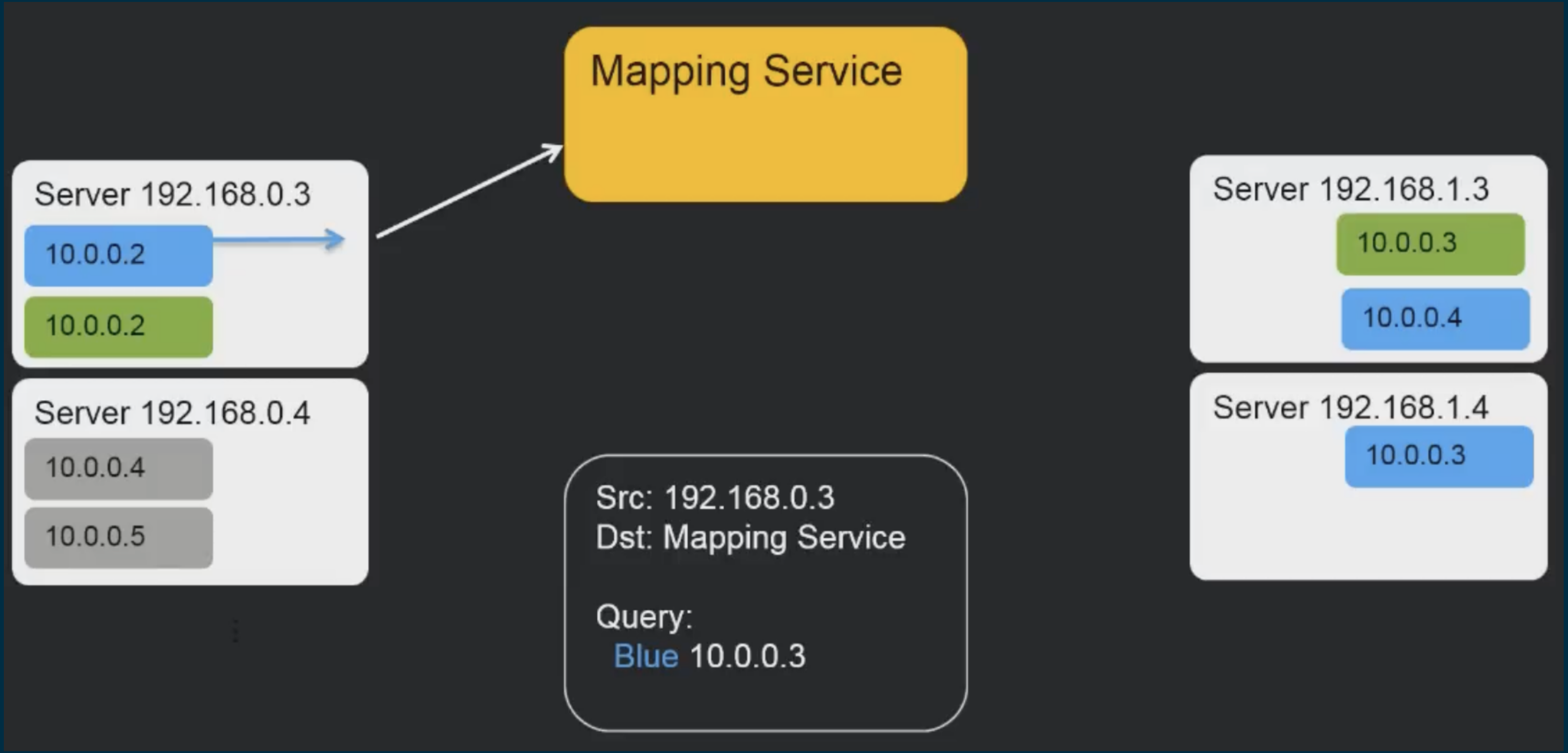


Ethernet Switch
10.0.0.2
10.0.0.3
L2 Src: MAC(10.0.0.2)
L2 Dst: MAC(10.0.0.3)
L3 Src: 10.0.0.2
L3 Dst: 10.0.0.3
ICMP/TCP/UDP/...

L2 - VPC

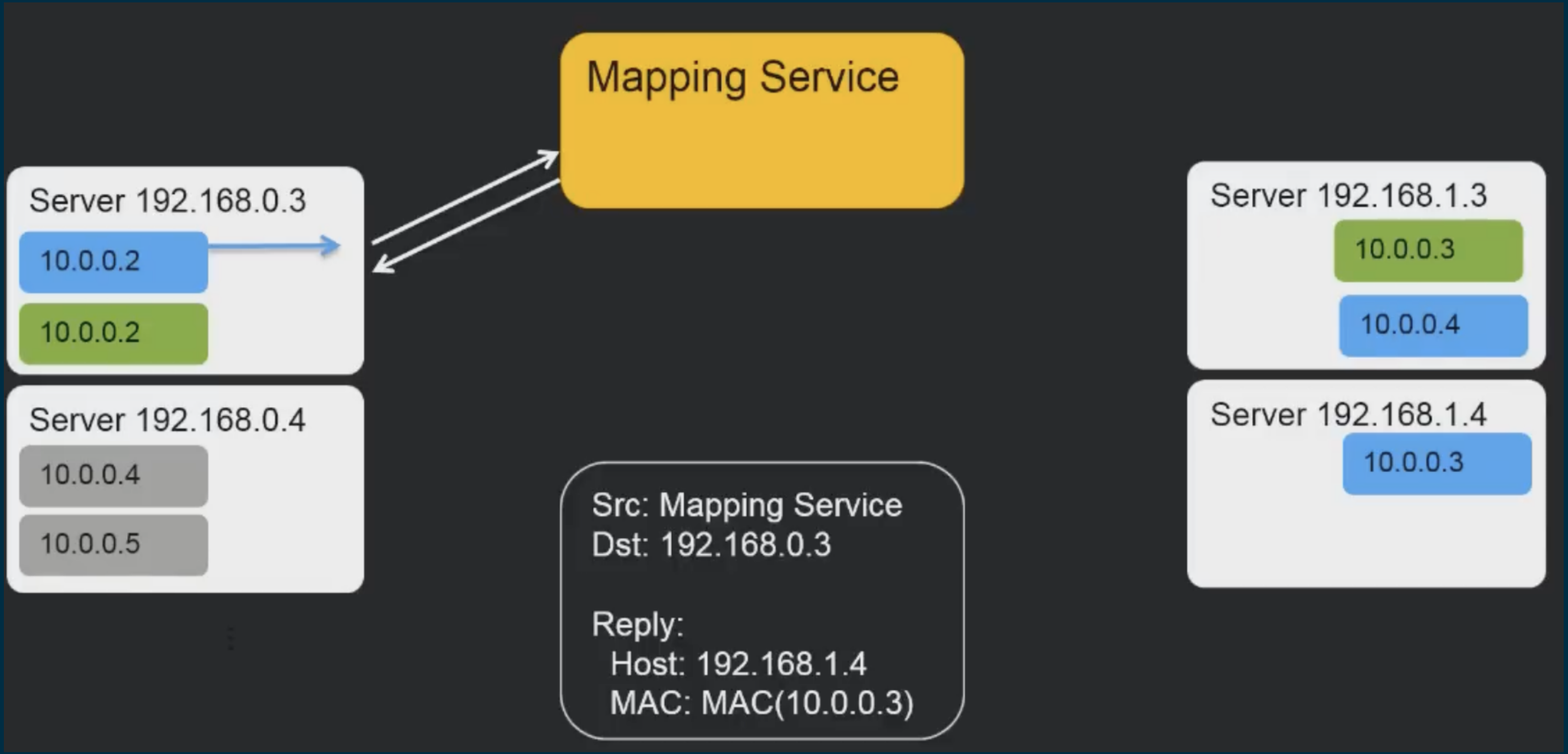

Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
L2 Src: MAC(10.0.0.2)
L2 Dst: ff:ff:ff:ff:ff:ff
ARP Who has
10.0.0.3?
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.0.3

L2 - VPC



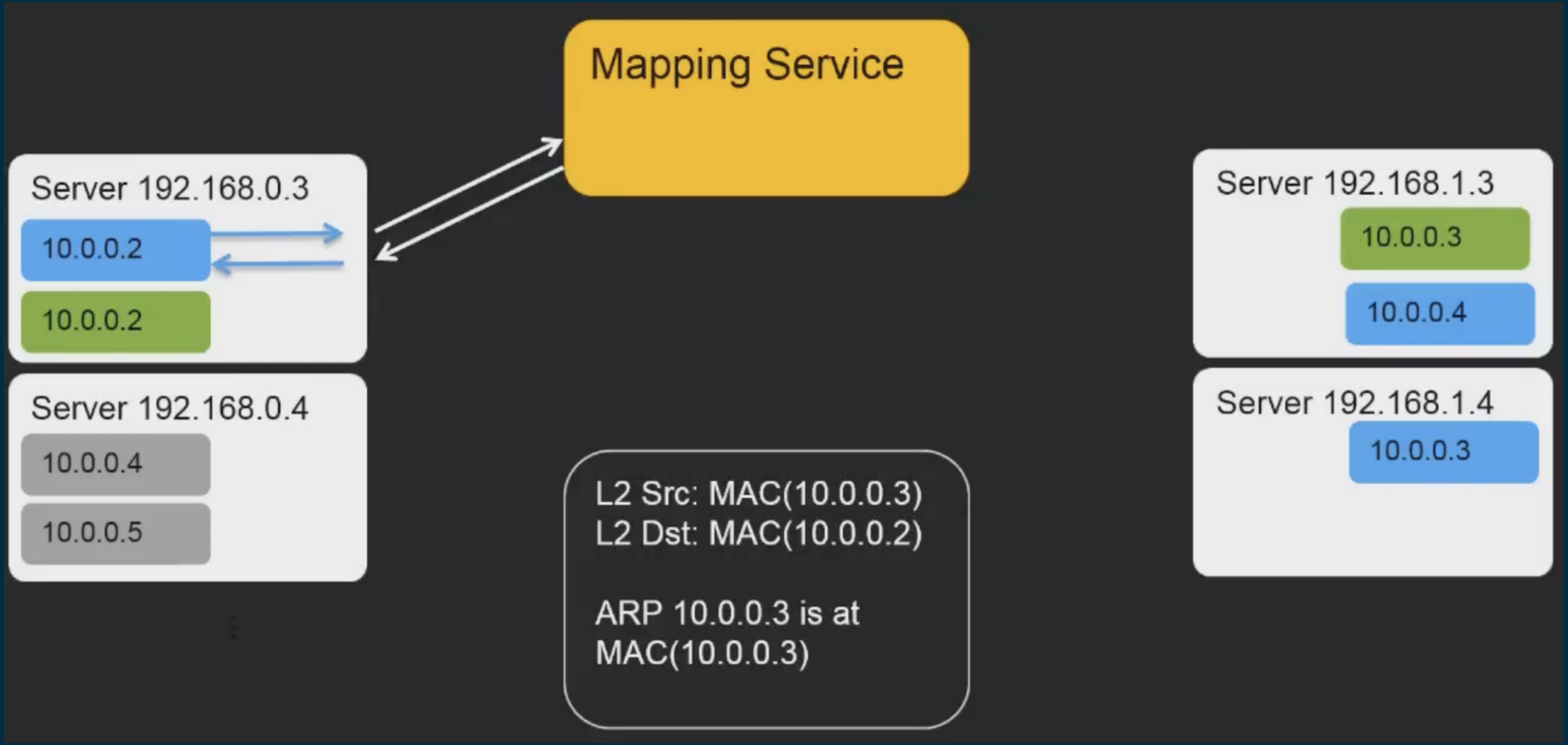
Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.0.3
Src: 192.168.0.3
Dst: Mapping Service
Query:
Blue 10.0.0.3

L2 - VPC

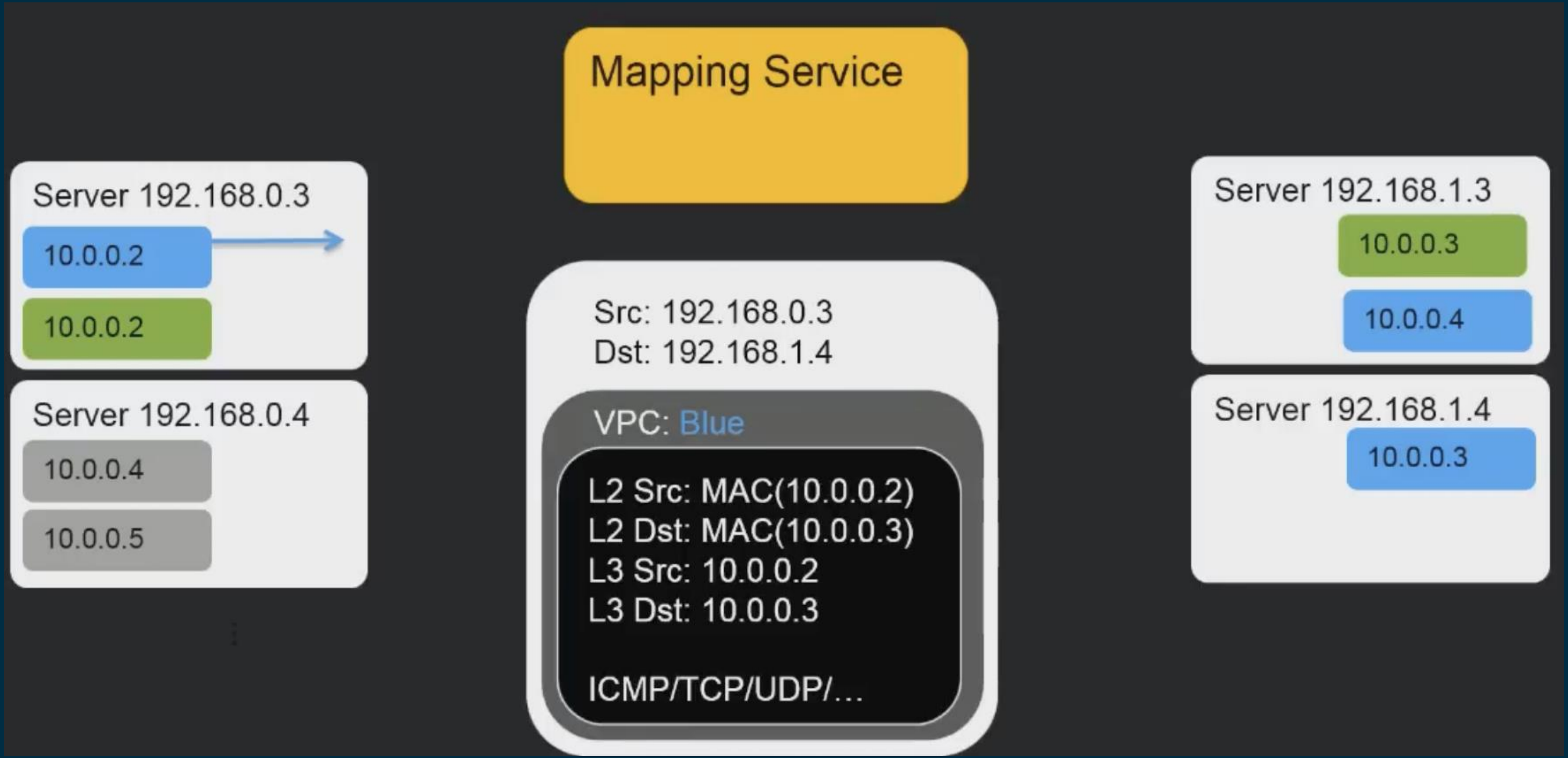


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.0.3
Src: Mapping Service
Dst: 192.168.0.3
Reply:
Host: 192.168.1.4
MAC: MAC(10.0.0.3)

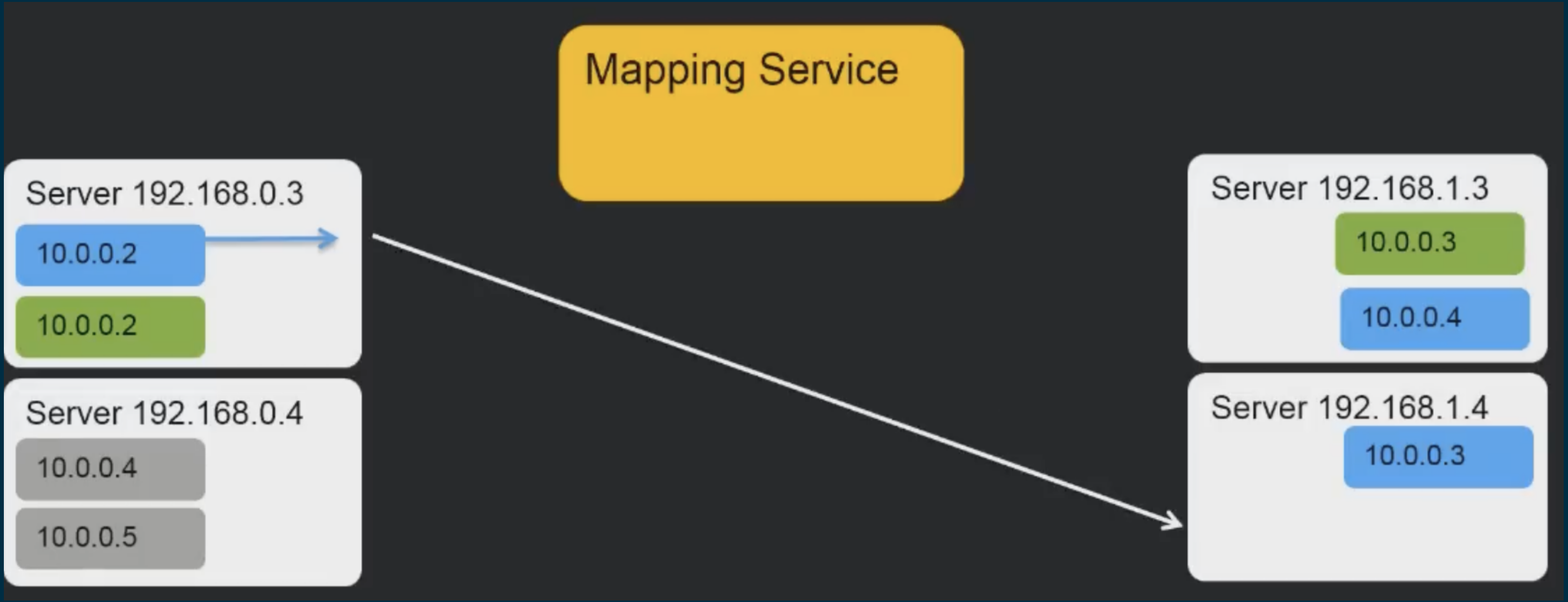
L2 - VPC


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.0.3
L2 Src: MAC(10.0.0.3)
L2 Dst: MAC(10.0.0.2)
ARP 10.0.0.3 is at
MAC(10.0.0.3)

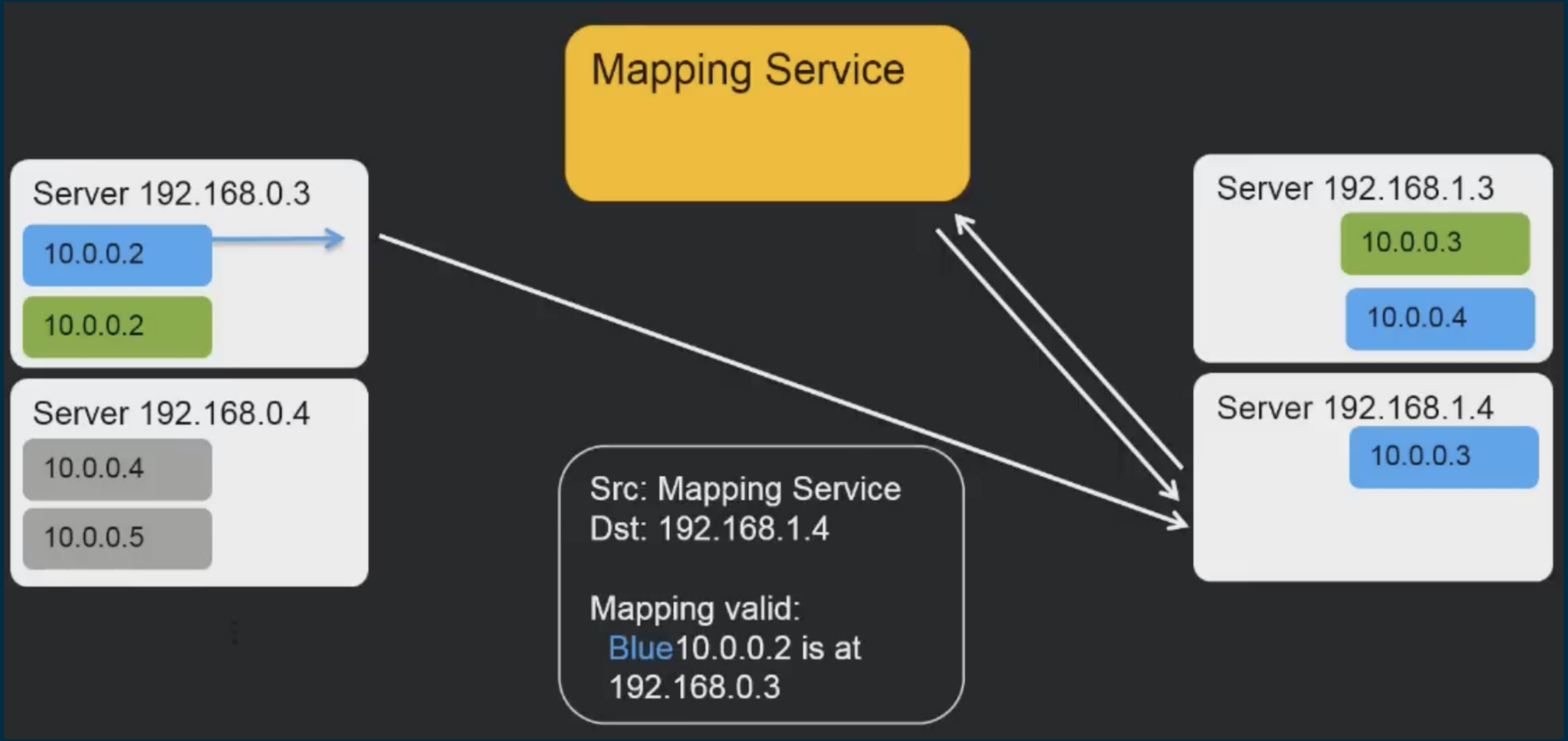
L2 - VPC


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Src: 192.168.0.3
Dst: 192.168.1.4
VPC: Blue
L2 Src: MAC(10.0.0.2)
L2 Dst: MAC(10.0.0.3)
L3 Src: 10.0.0.2
L3 Dst: 10.0.0.3
ICMP/TCP/UDP/...
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.0.3

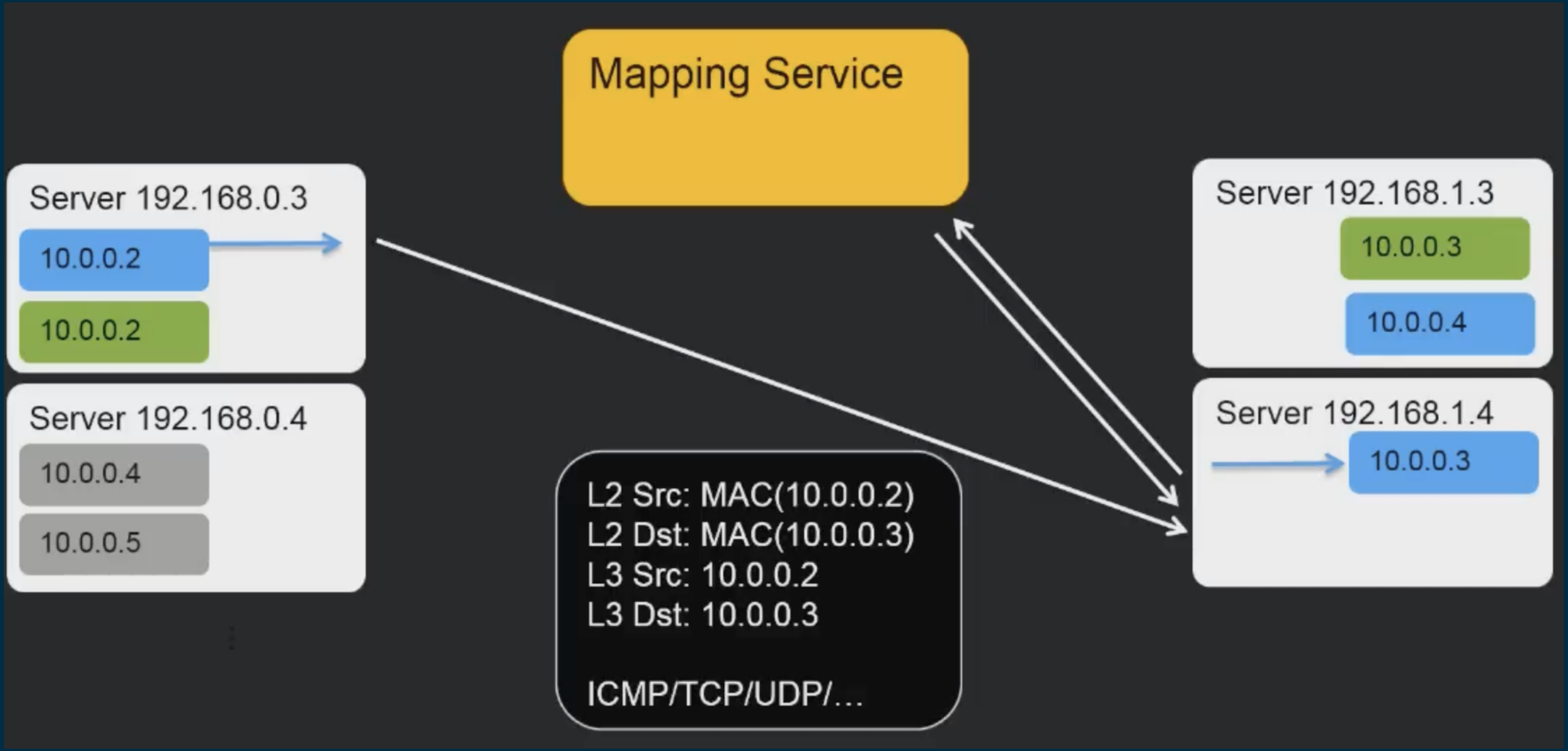
L2 - VPC


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.0.3

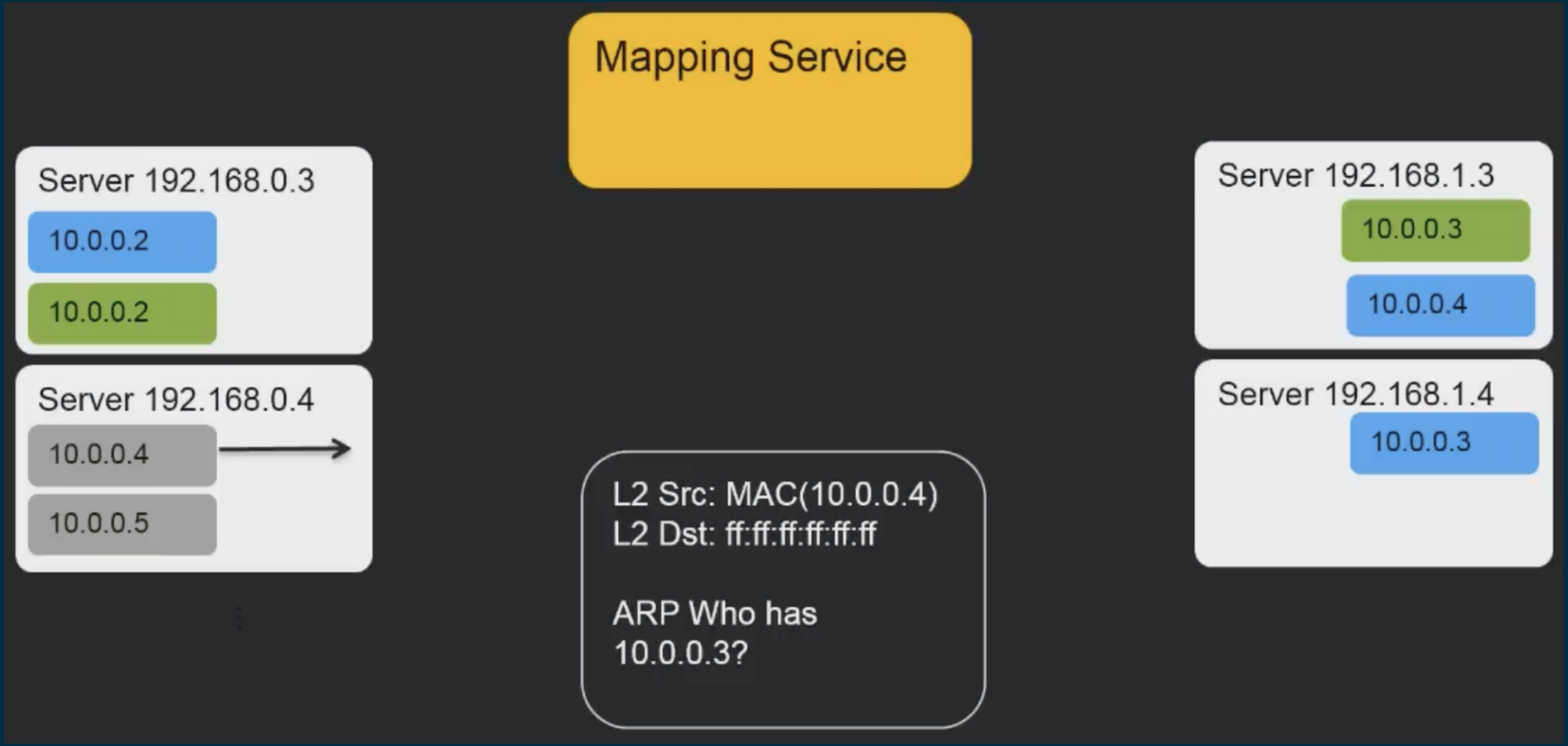
L2 - VPC


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.0.3
Src: Mapping Service
Dst: 192.168.1.4
Mapping valid:
Blue10.0.0.2 is at
192.168.0.3

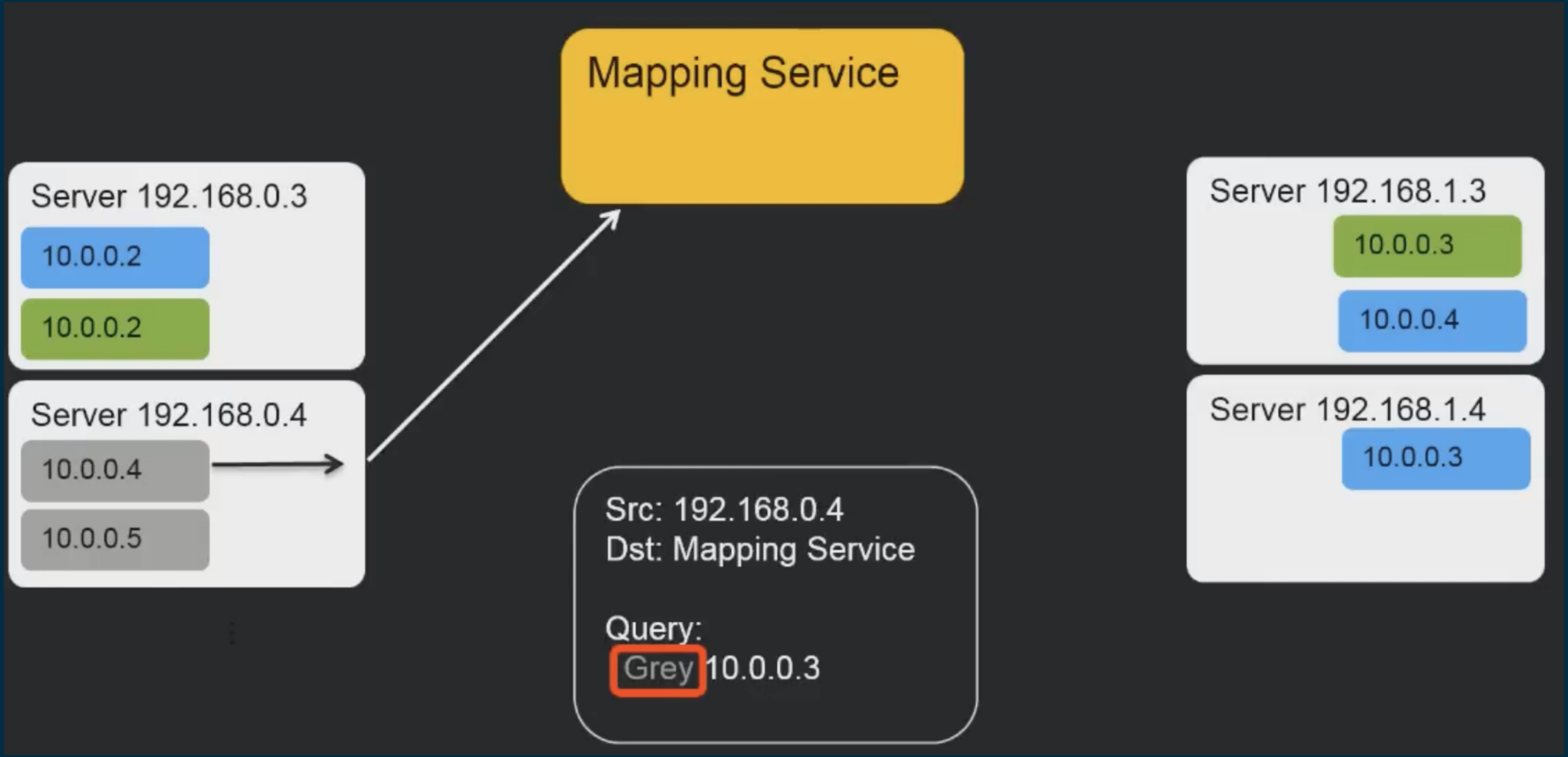
L2 - VPC


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.0.3
L2 Src: MAC(10.0.0.2)
L2 Dst: MAC(10.0.0.3)
L3 Src: 10.0.0.2
L3 Dst: 10.0.0.3
ICMP/TCP/UDP/…

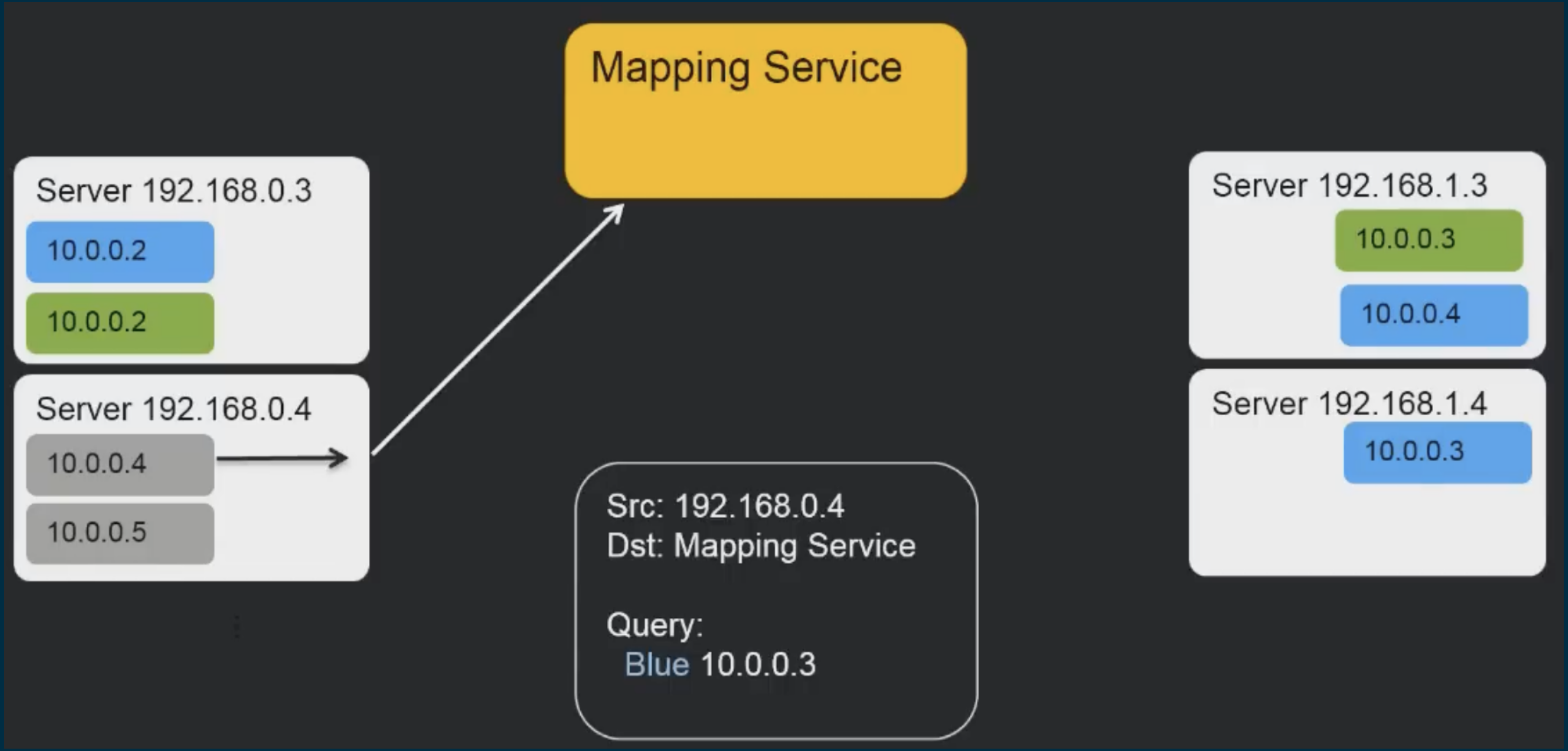
VPC Isolation



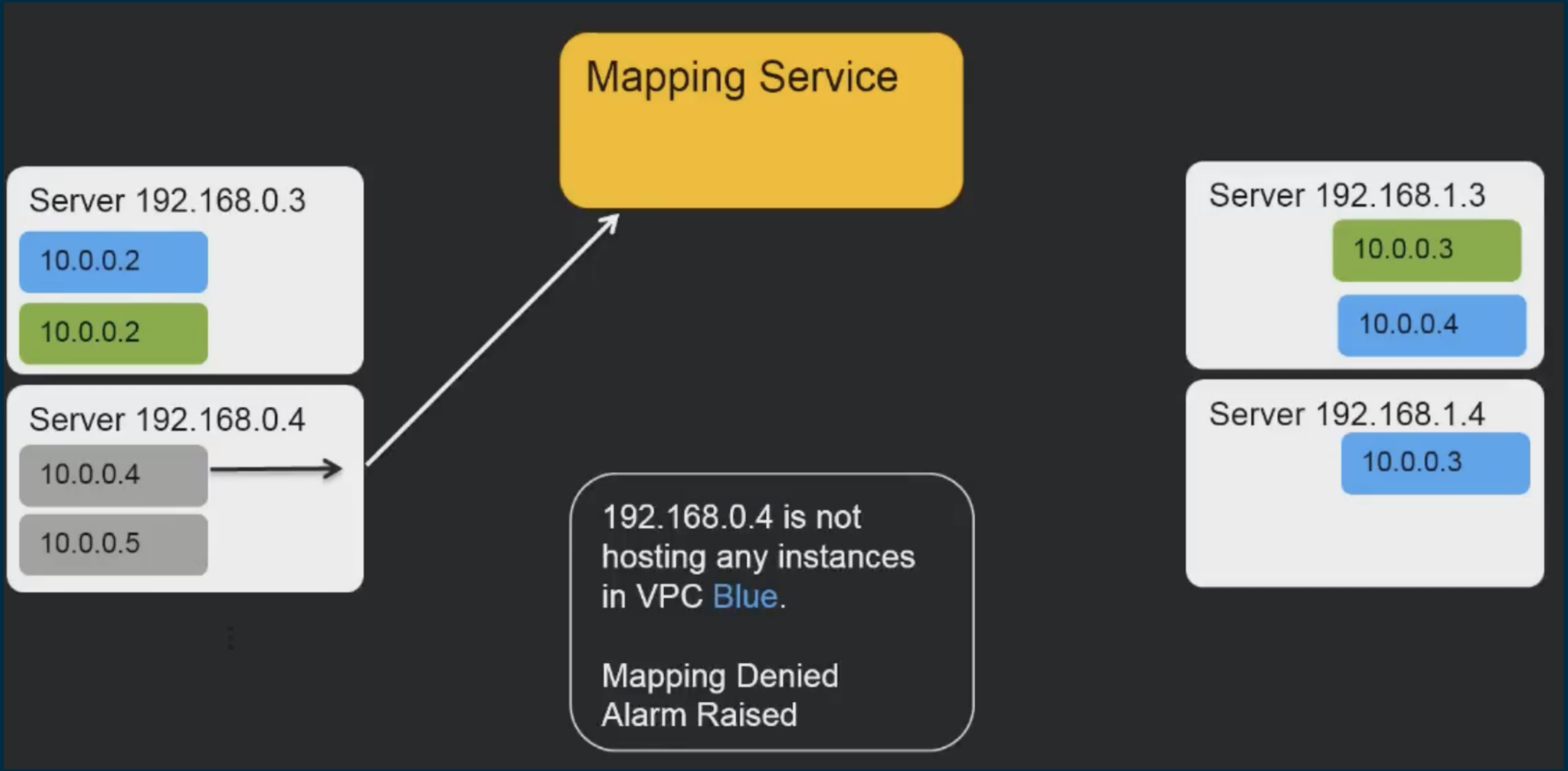
VPC Isolation



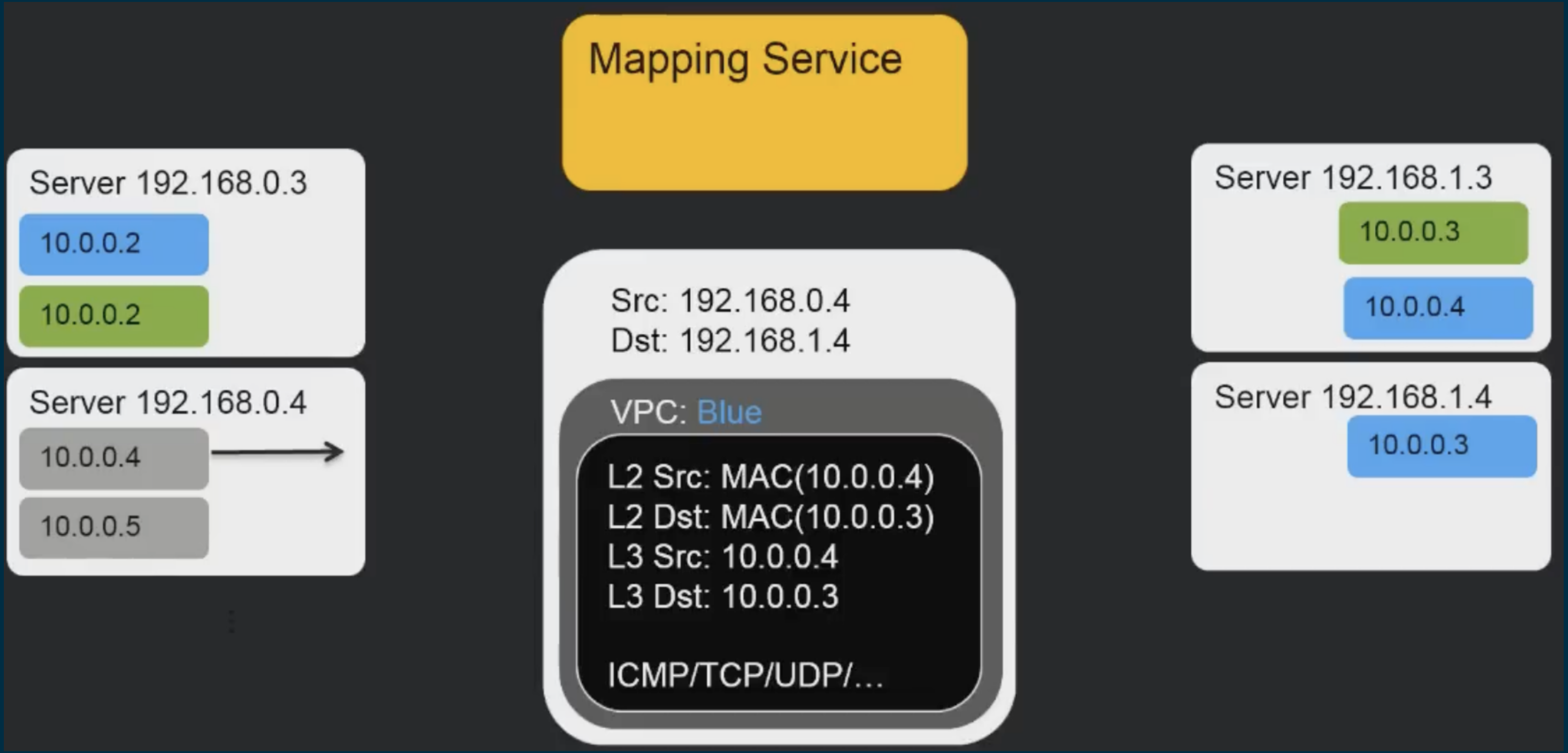
VPC Isolation



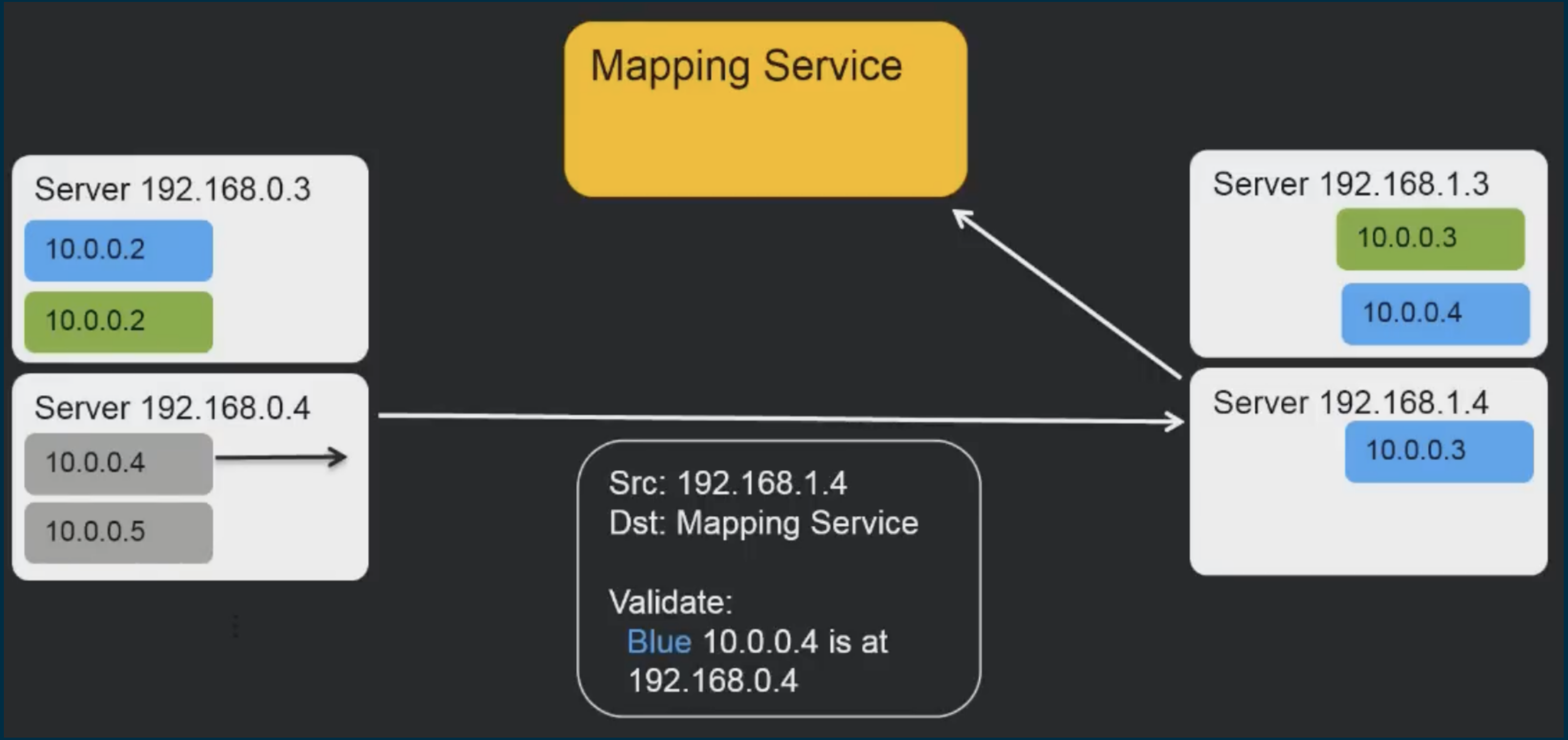
VPC Isolation



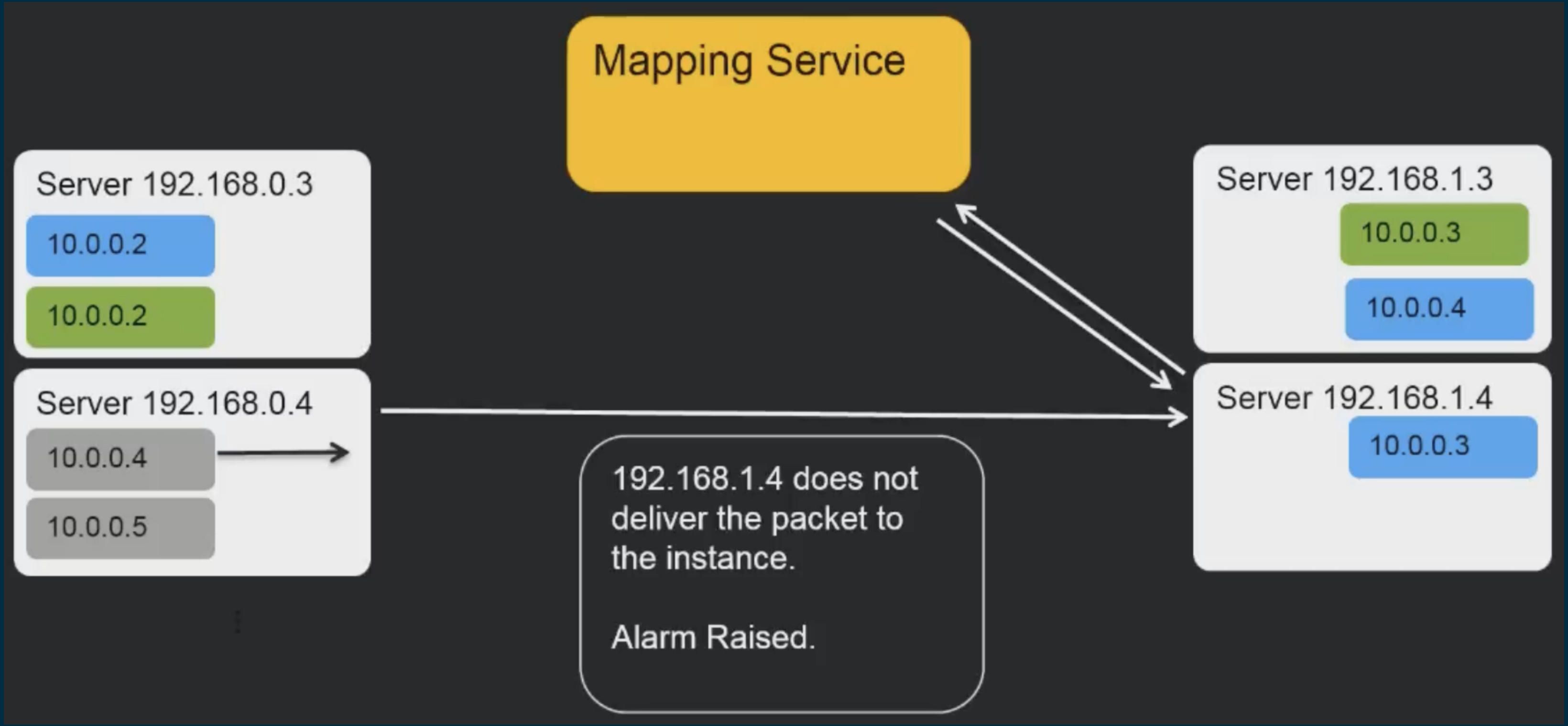
VPC Isolation



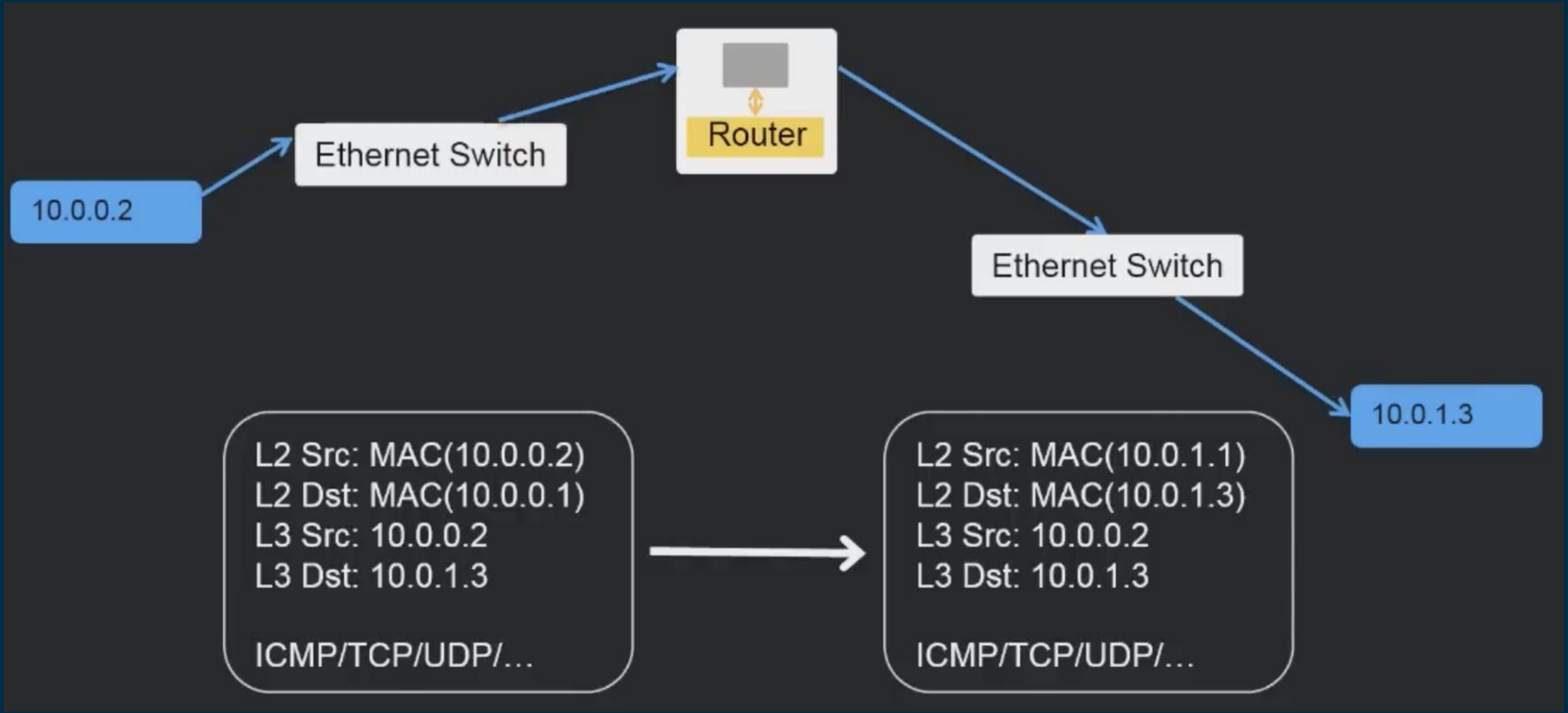
VPC Isolation



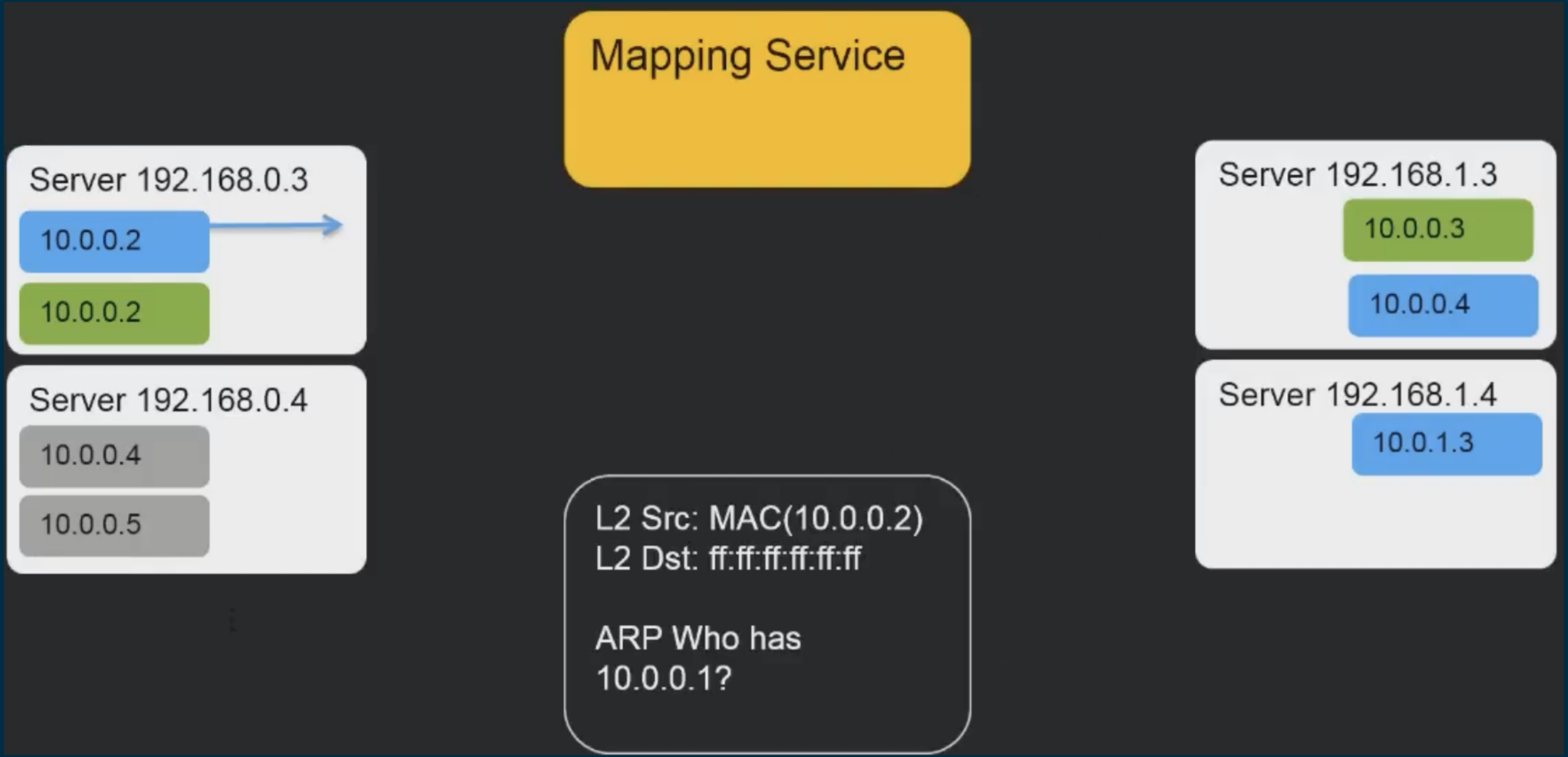
VPC Isolation



L3 – IP Routing

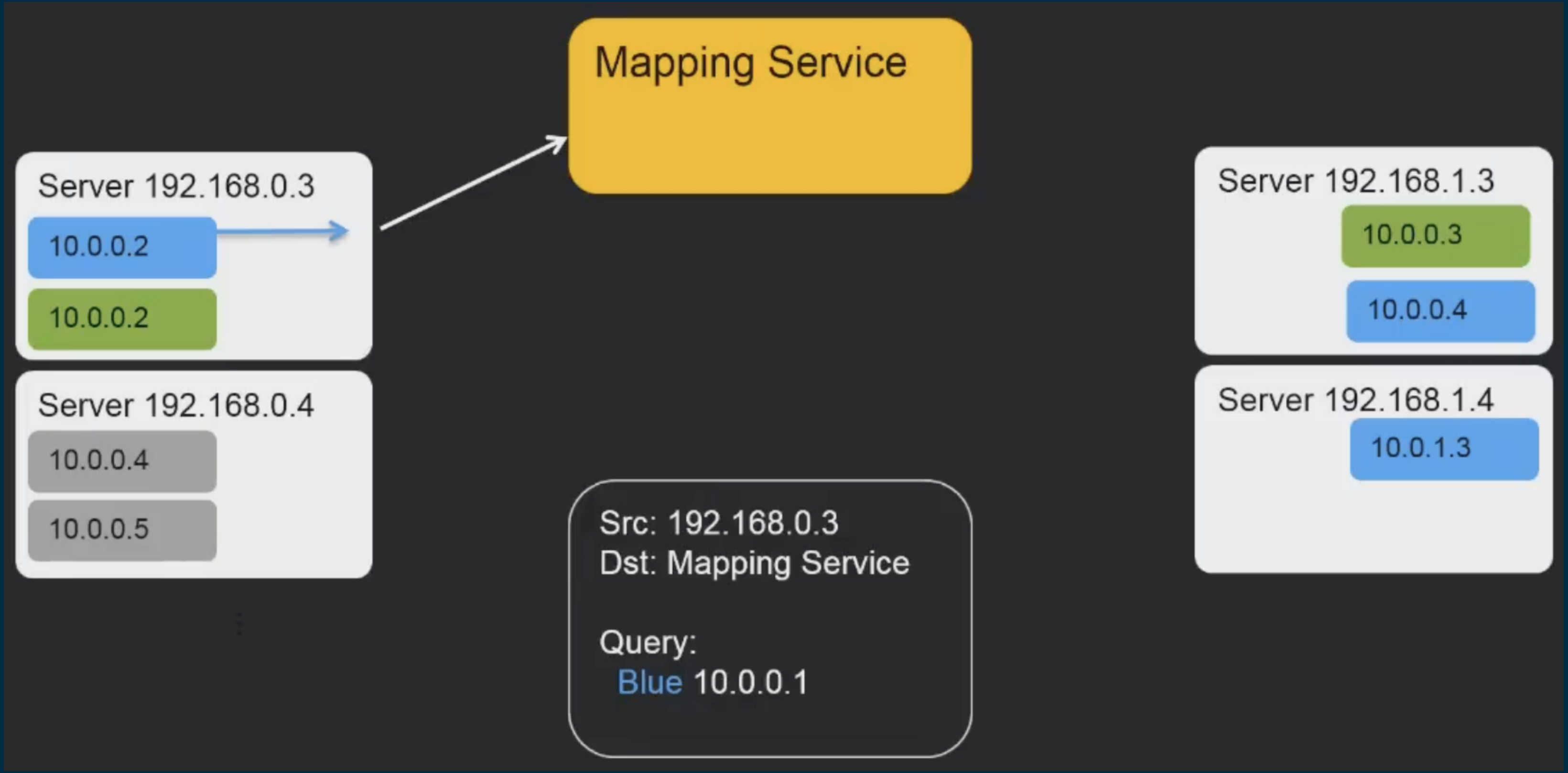


L3 – VPC



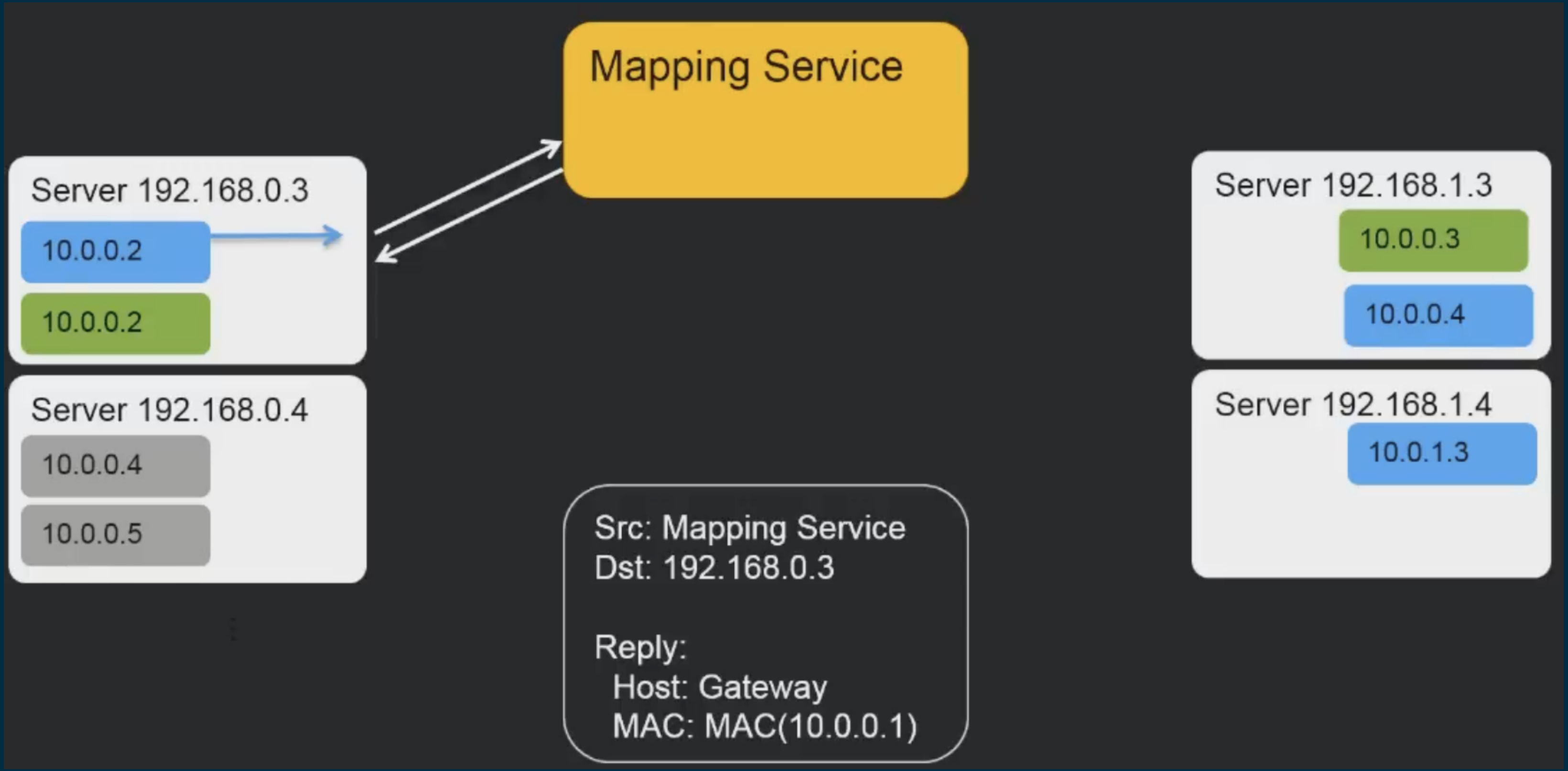
Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.1.3
L2 Src: MAC(10.0.0.2)
L2 Dst: ff:ff:ff:ff:ff:ff
ARP Who has
10.0.0.1?

L3 – VPC

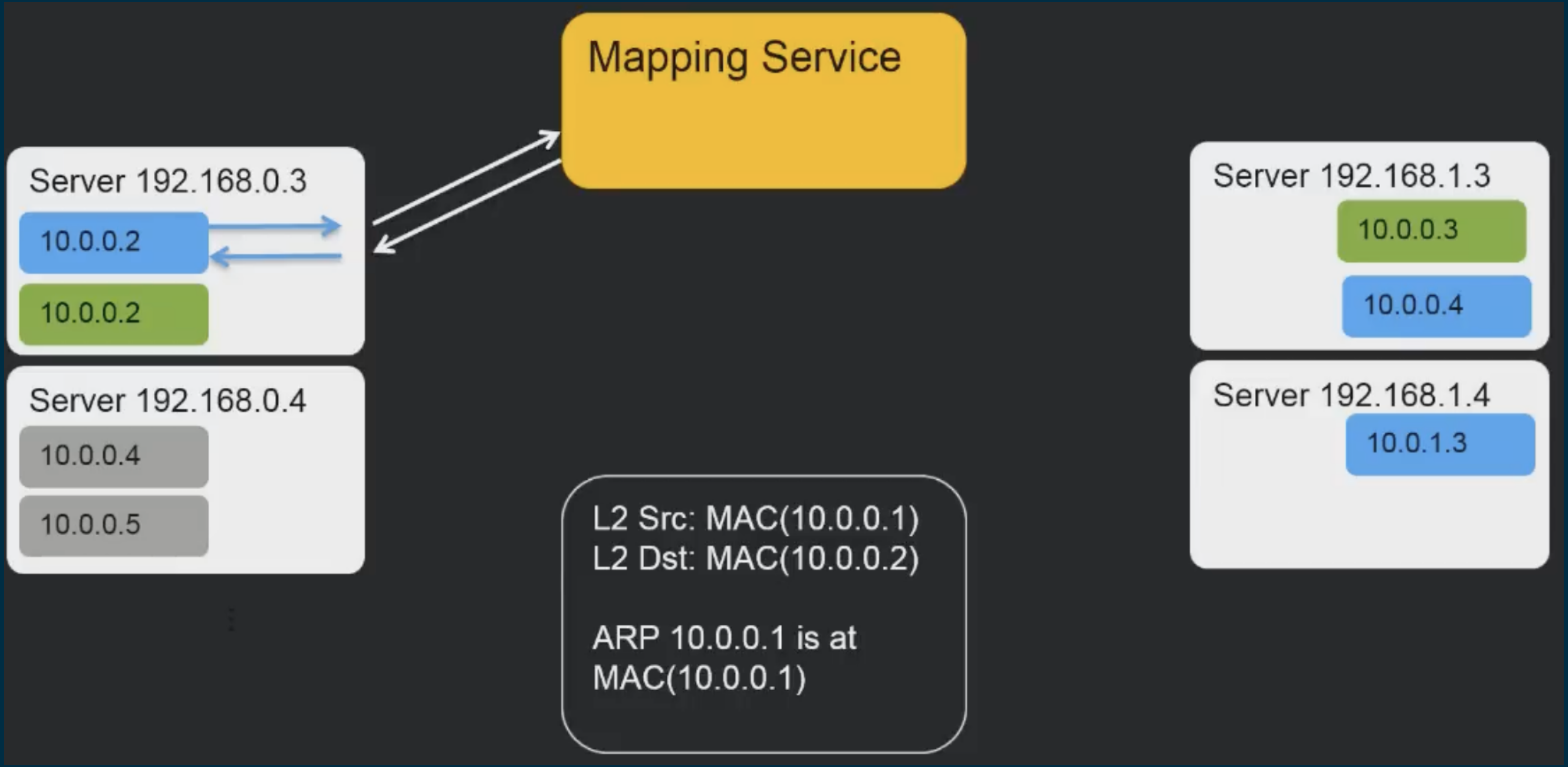


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.1.3
Src: 192.168.0.3
Dst: Mapping Service
Query:
Blue 10.0.0.1

L3 – VPC

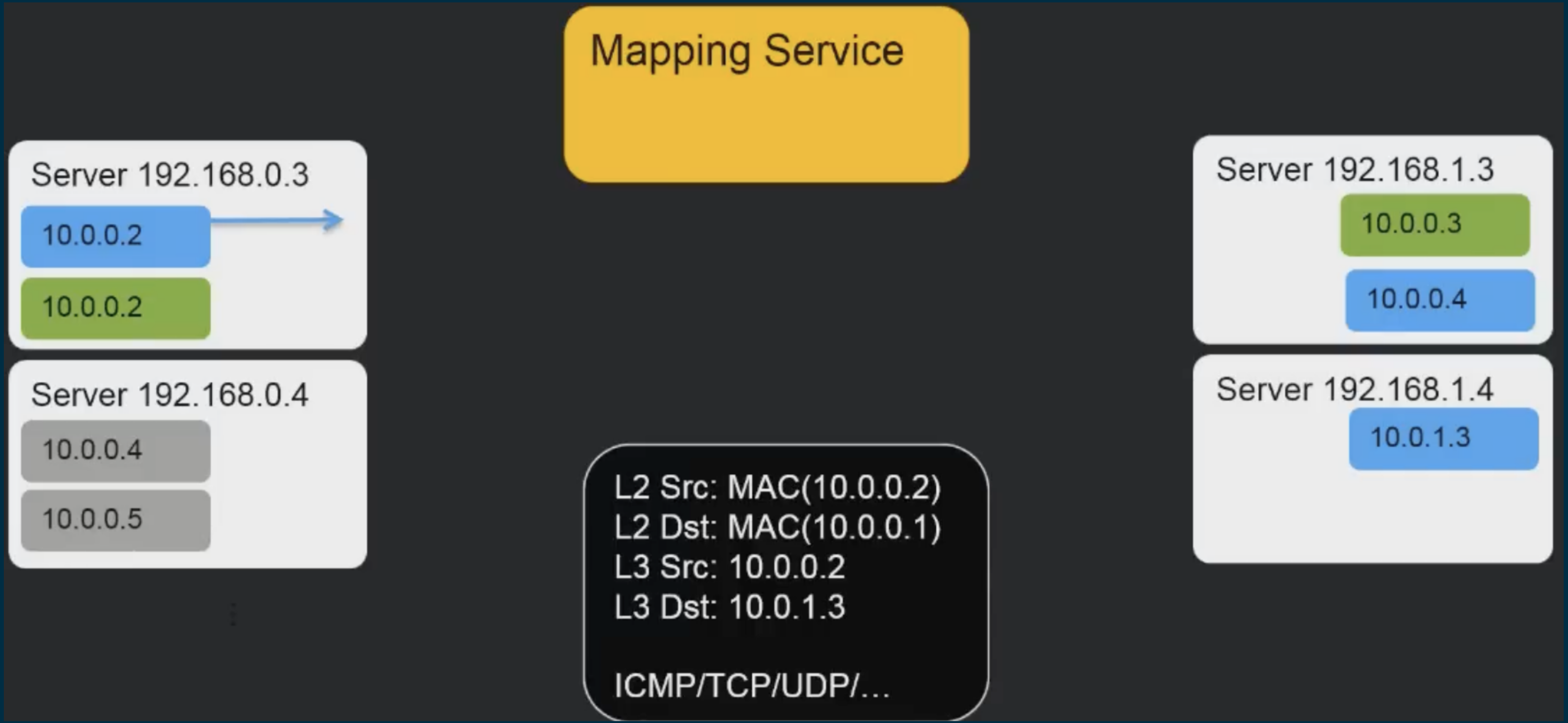

Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.1.3
Src: Mapping Service
Dst: 192.168.0.3
Reply:
Host: Gateway
MAC: MAC(10.0.0.1)

L3 – VPC

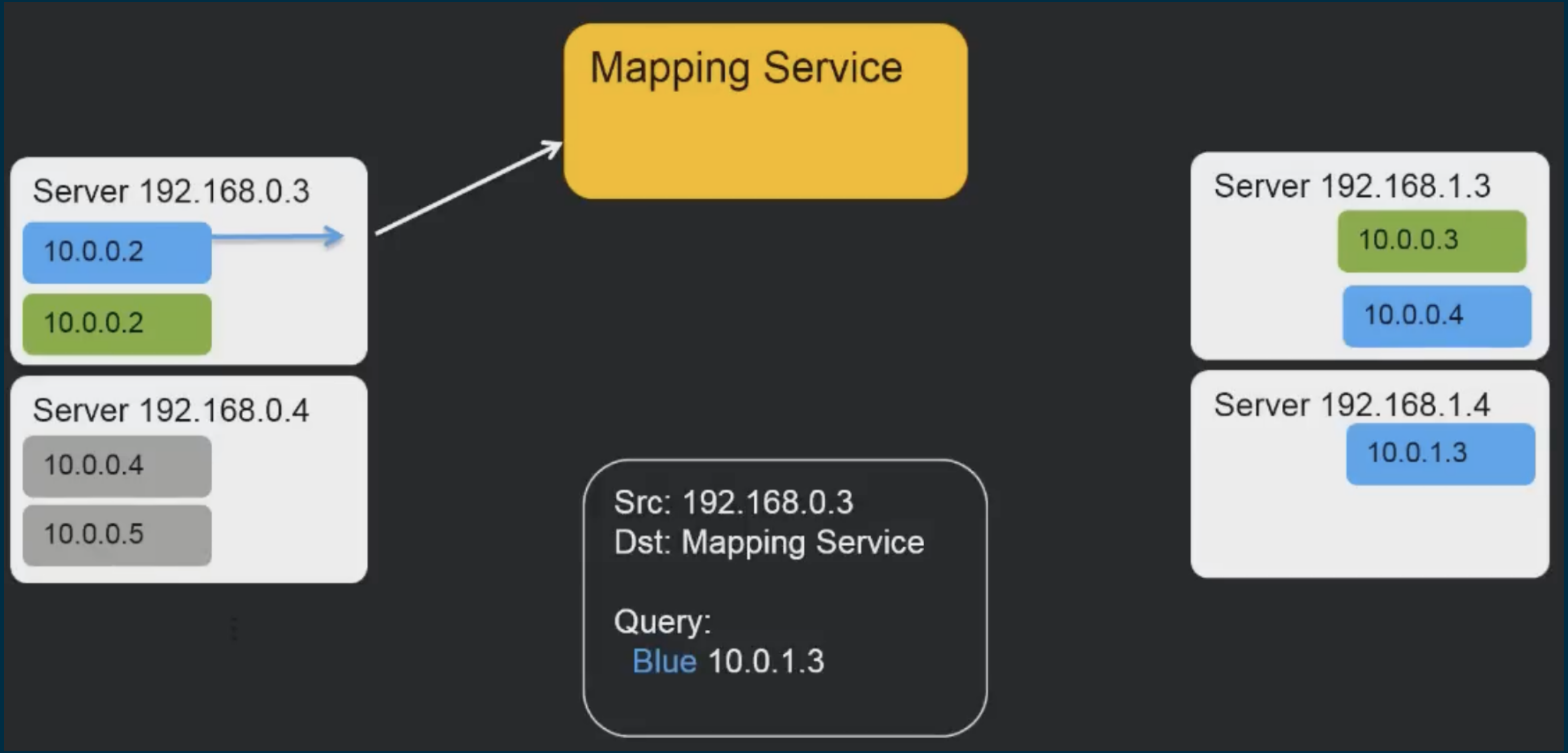


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.1.3
L2 Src: MAC(10.0.0.1)
L2 Dst: MAC(10.0.0.2)
ARP 10.0.0.1 is at
MAC(10.0.0.1)

L3 – VPC

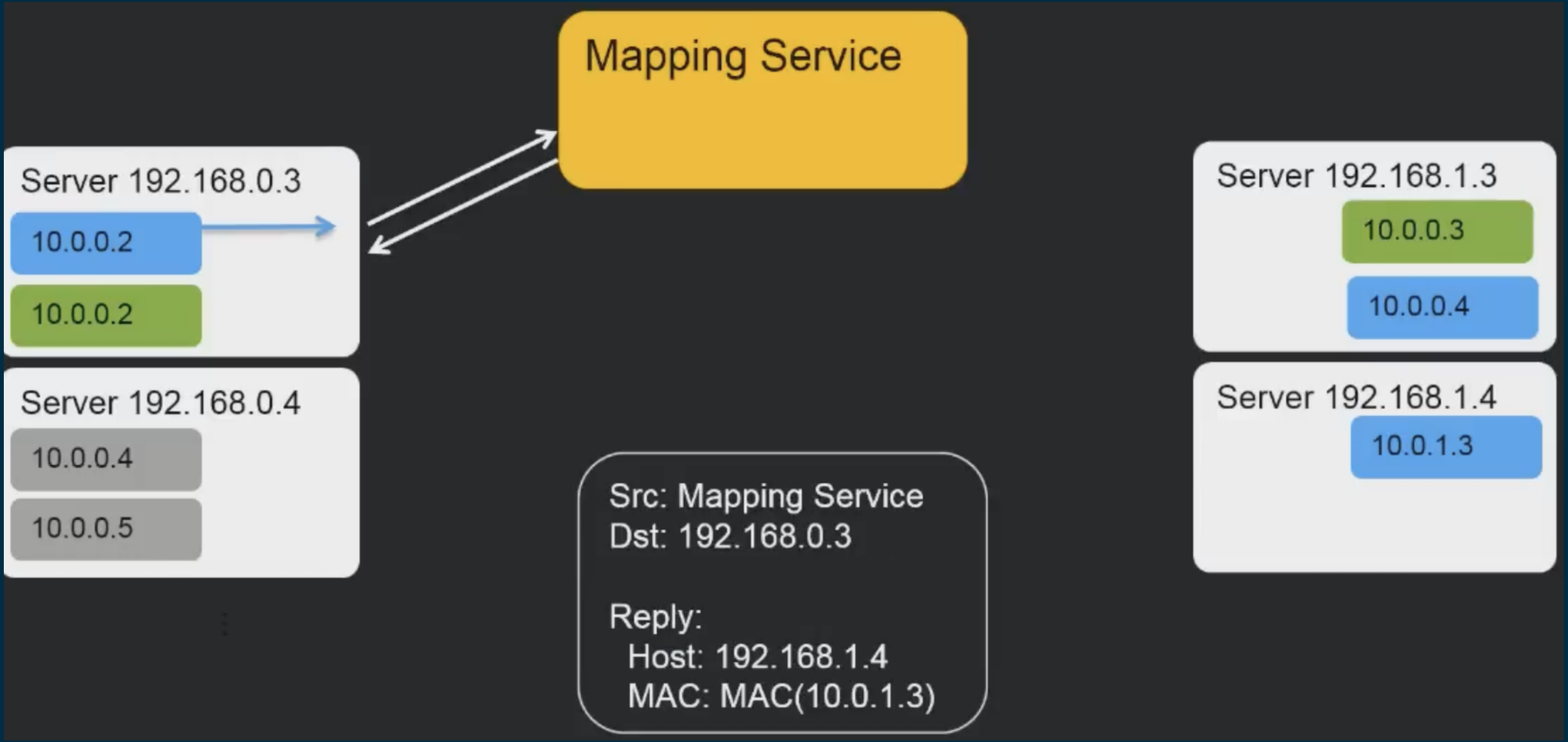

Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.1.3
L2 Src: MAC(10.0.0.2)
L2 Dst: MAC(10.0.0.1)
L3 Src: 10.0.0.2
L3 Dst: 10.0.1.3
ICMP/TCP/UDP/...

L3 – VPC

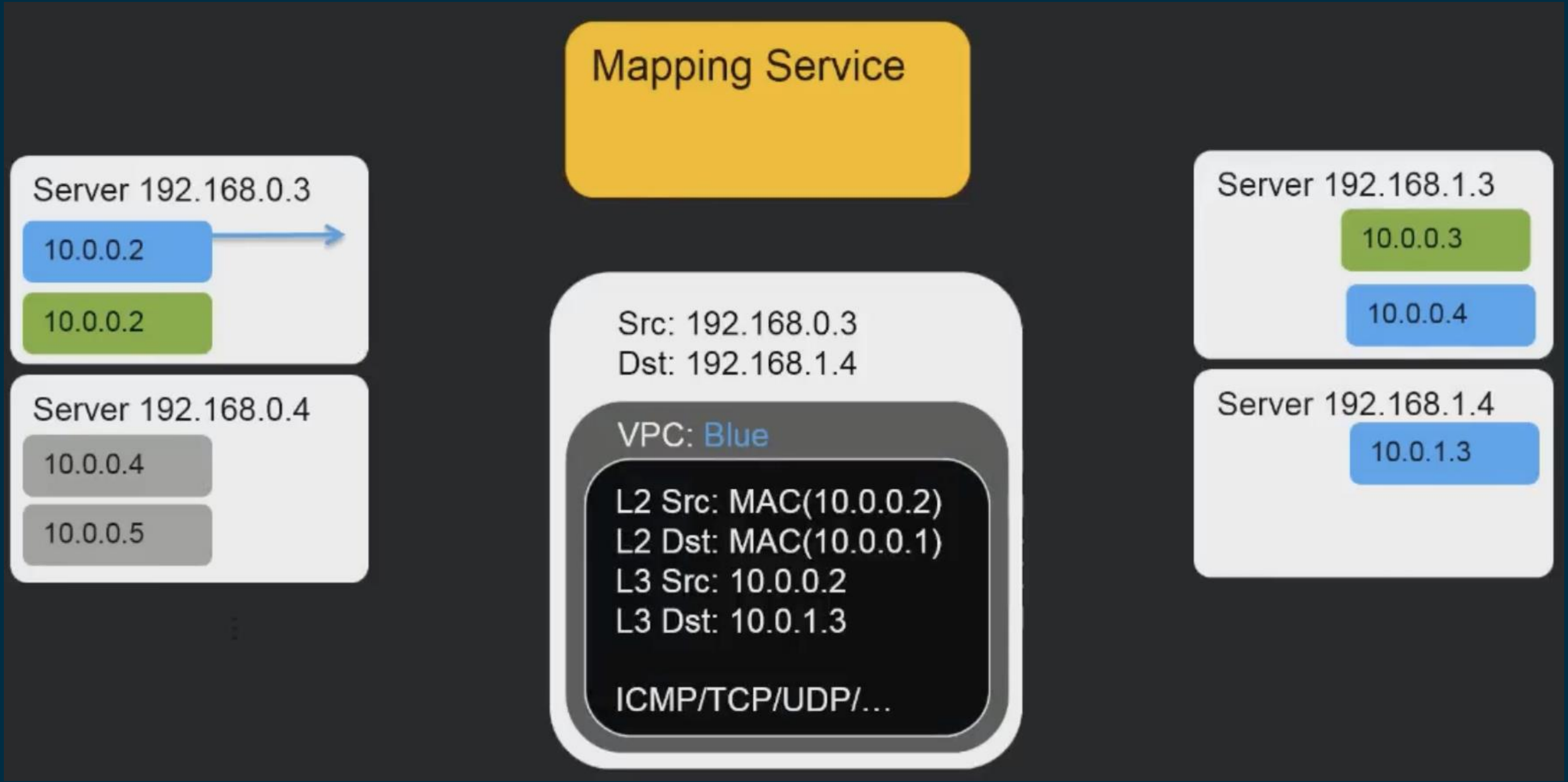


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.1.3
Src: 192.168.0.3
Dst: Mapping Service
Query:
Blue 10.0.1.3

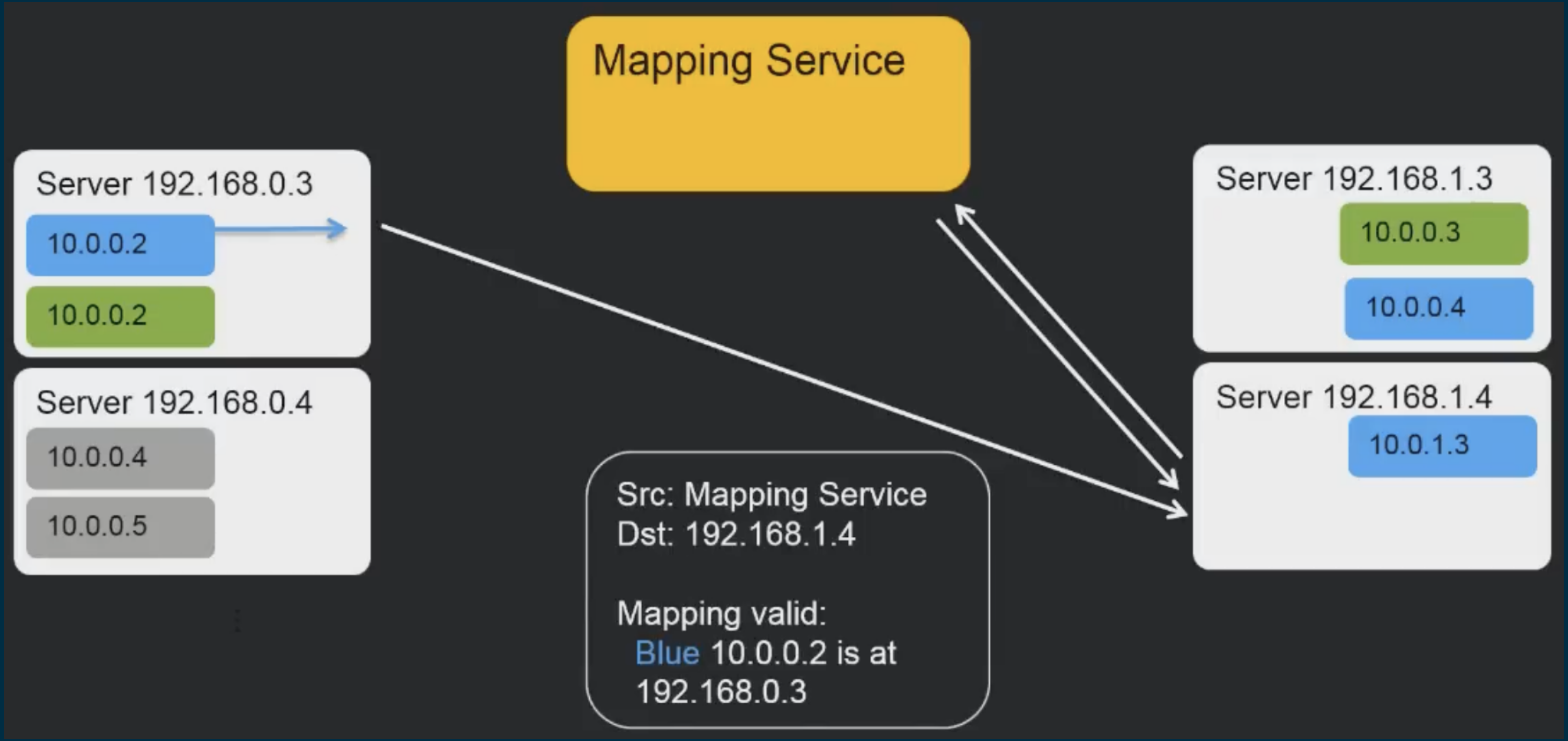
L3 – VPC


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.1.3
Src: Mapping Service
Dst: 192.168.0.3
Reply:
Host: 192.168.1.4
MAC: MAC(10.0.1.3)

L3 – VPC

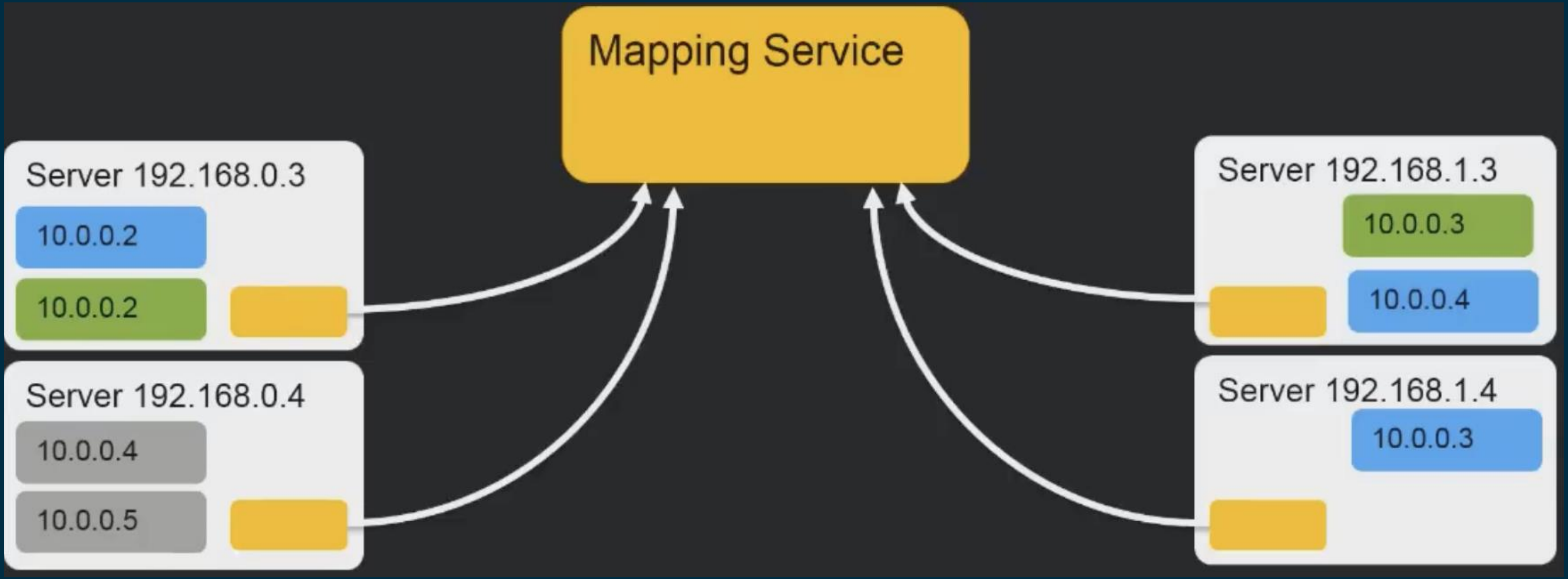

Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Src: 192.168.0.3
Dst: 192.168.1.4
VPC: Blue
L2 Src: MAC(10.0.0.2)
L2 Dst: MAC(10.0.0.1)
L3 Src: 10.0.0.2
L3 Dst: 10.0.1.3
ICMP/TCP/UDP/...
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.1.3

L3 – VPC

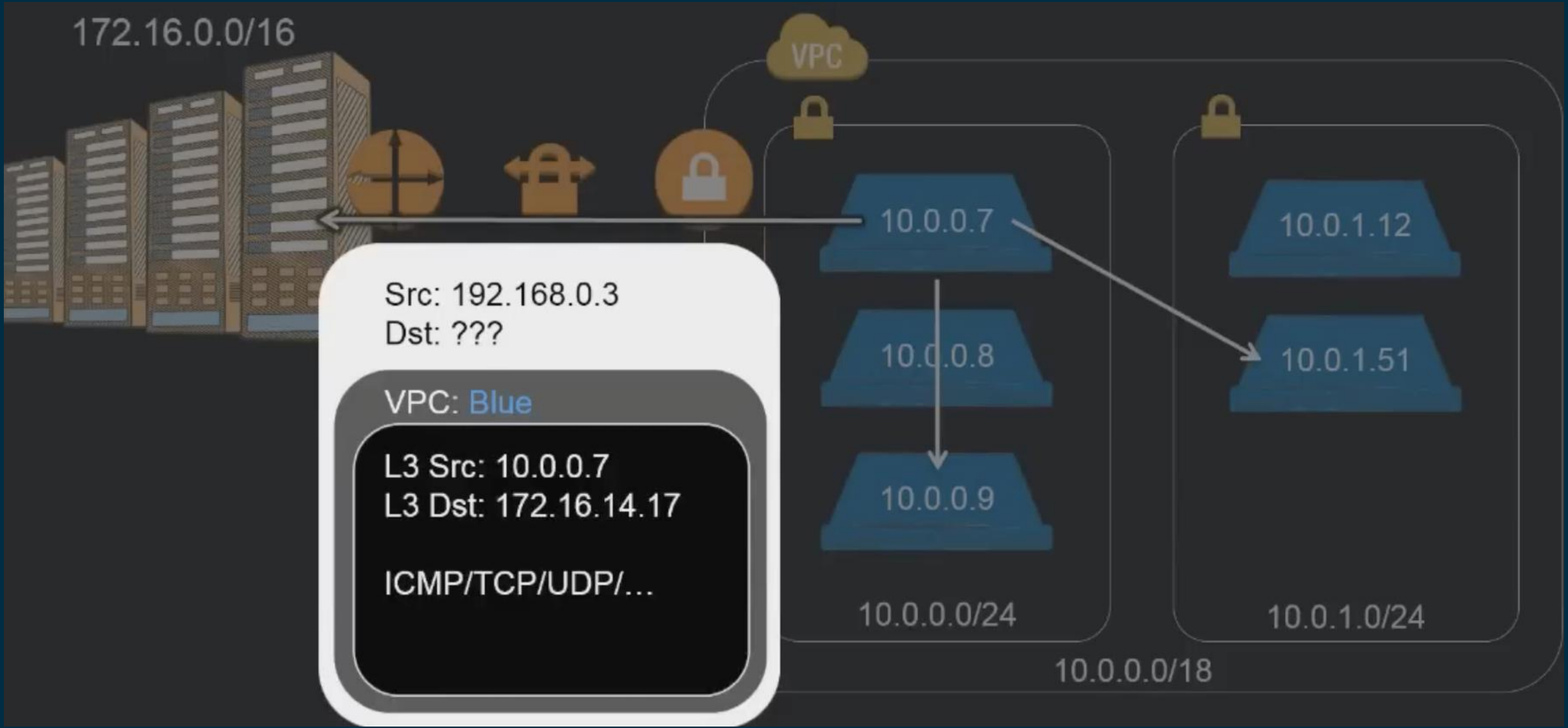


Mapping Service
Server 192.168.0.3
10.0.0.2
10.0.0.2
Server 192.168.0.4
10.0.0.4
10.0.0.5
Server 192.168.1.3
10.0.0.3
10.0.0.4
Server 192.168.1.4
10.0.1.3
Src: Mapping Service
Dst: 192.168.1.4
Mapping valid:
Blue 10.0.0.2 is at
192.168.0.3

Caching Mapping Service



On-premises Connectivity



Edges (Blackfoot)

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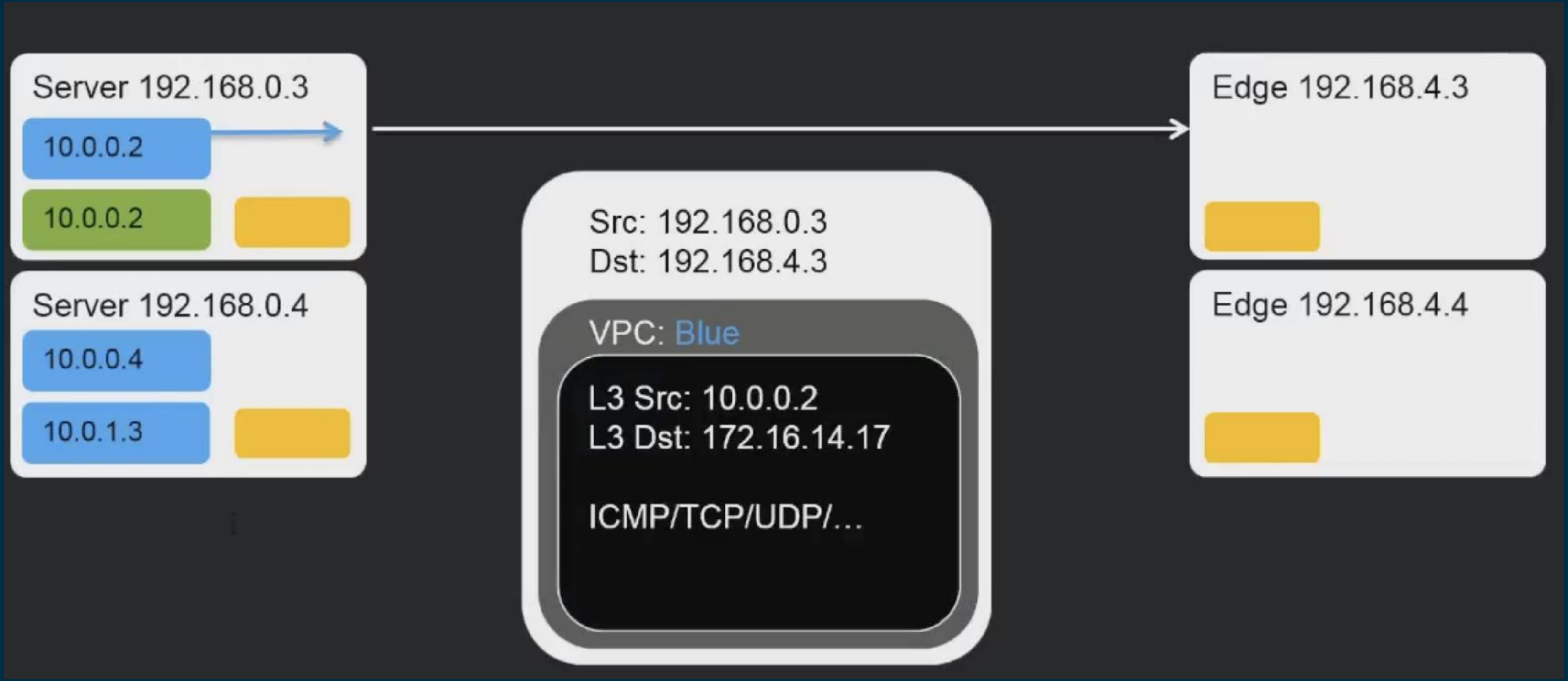
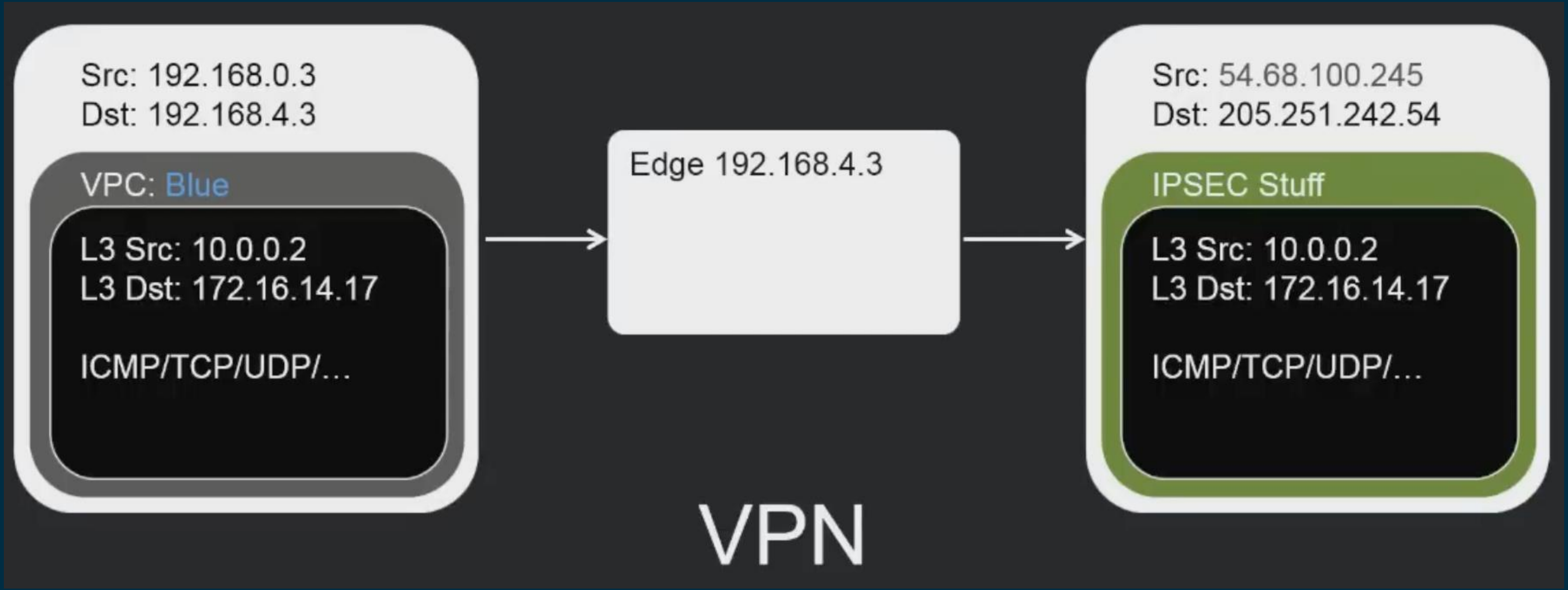
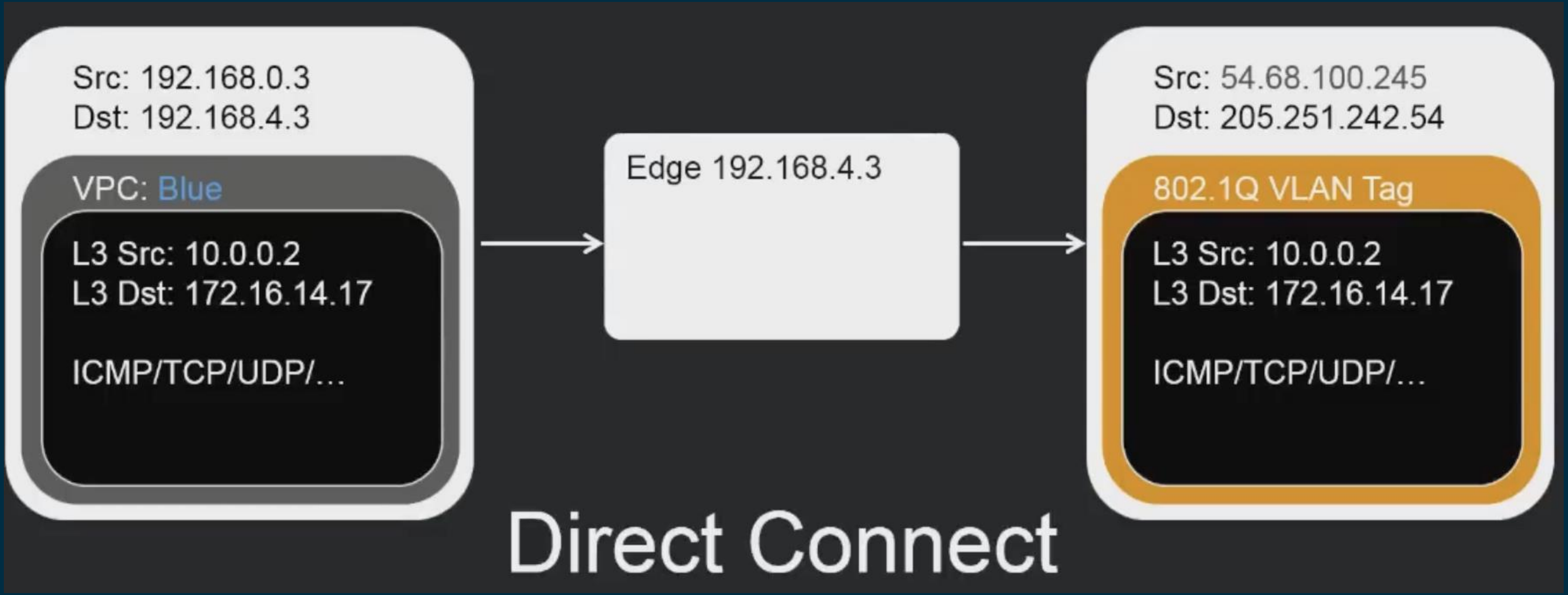


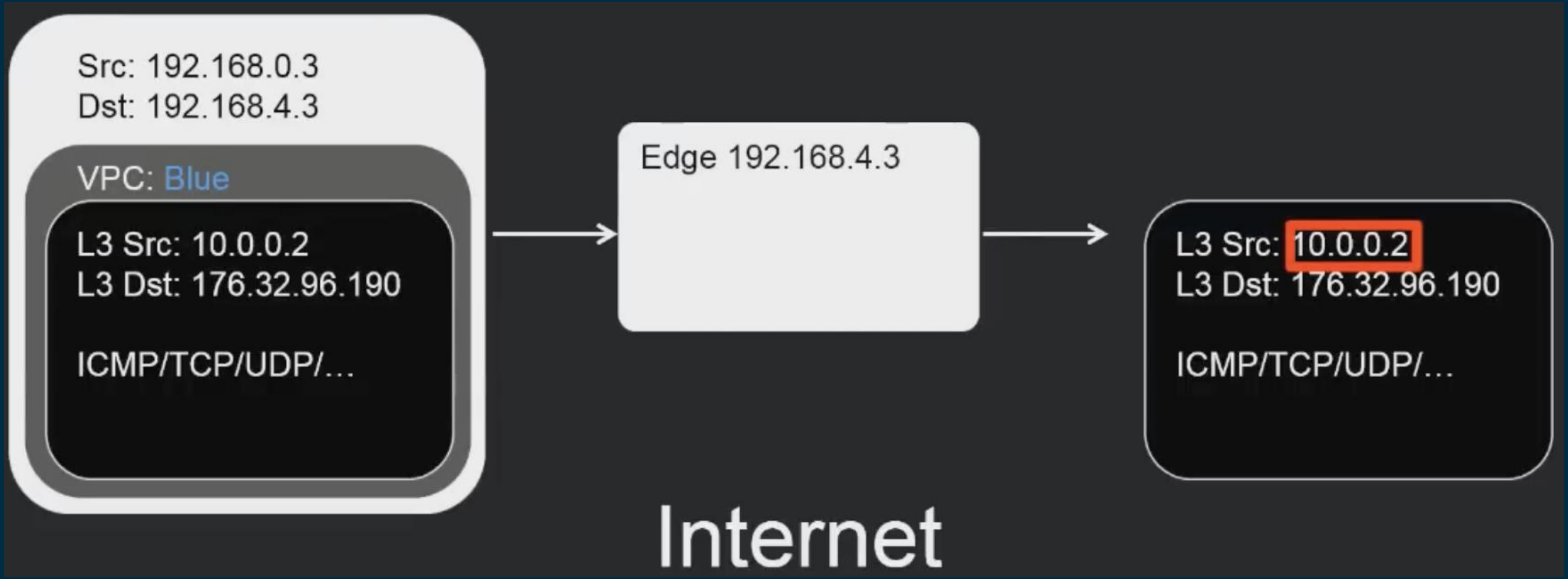
Edges (Blackfoot)



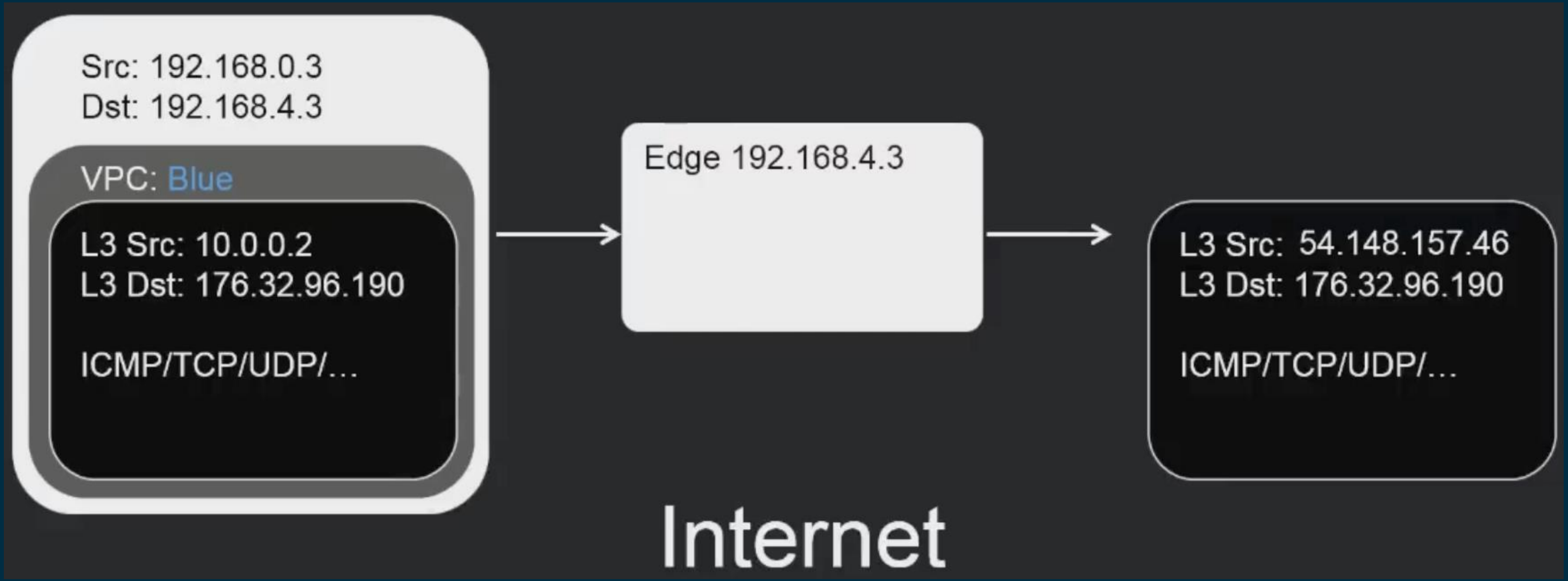
Edges (Blackfoot)



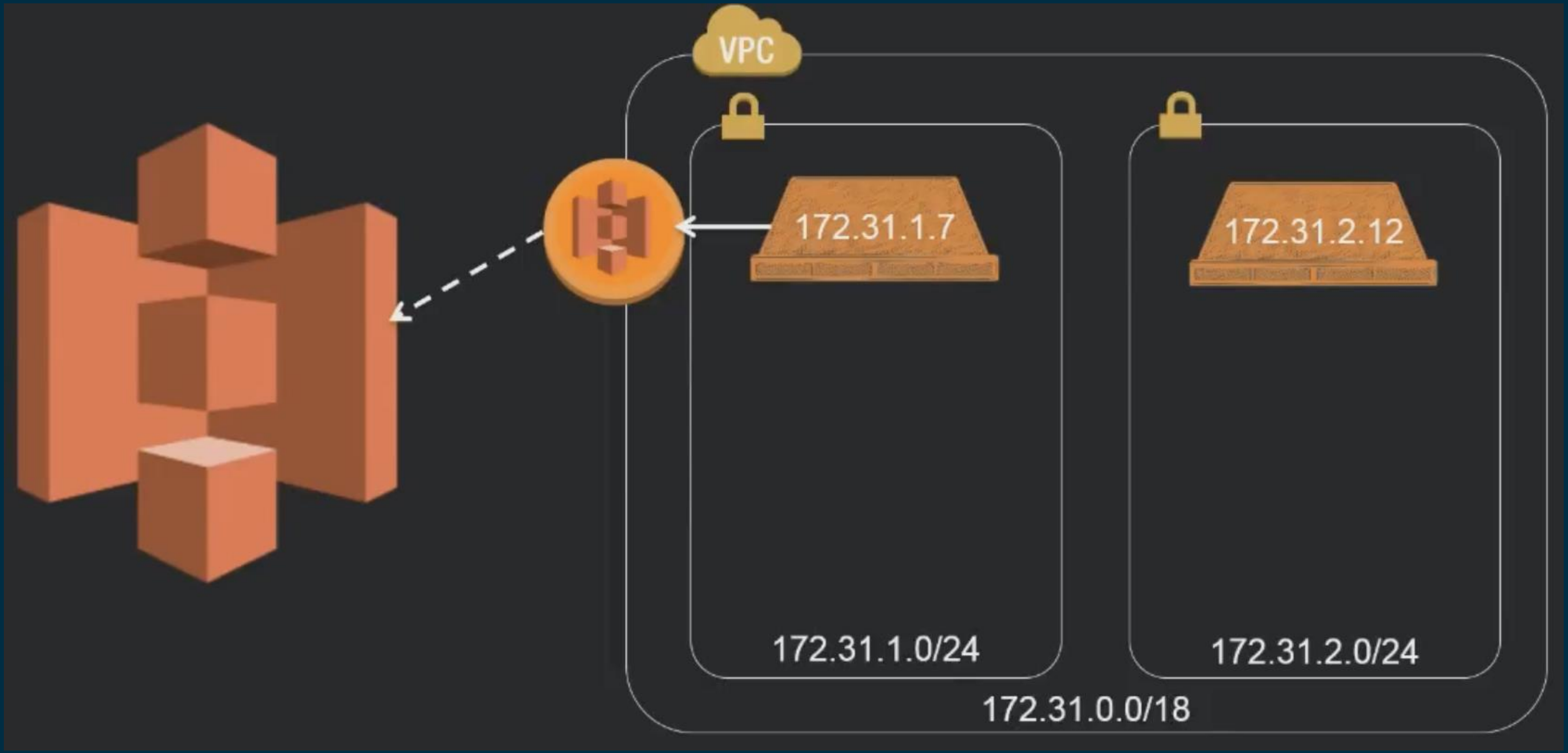
Edges (Blackfoot)



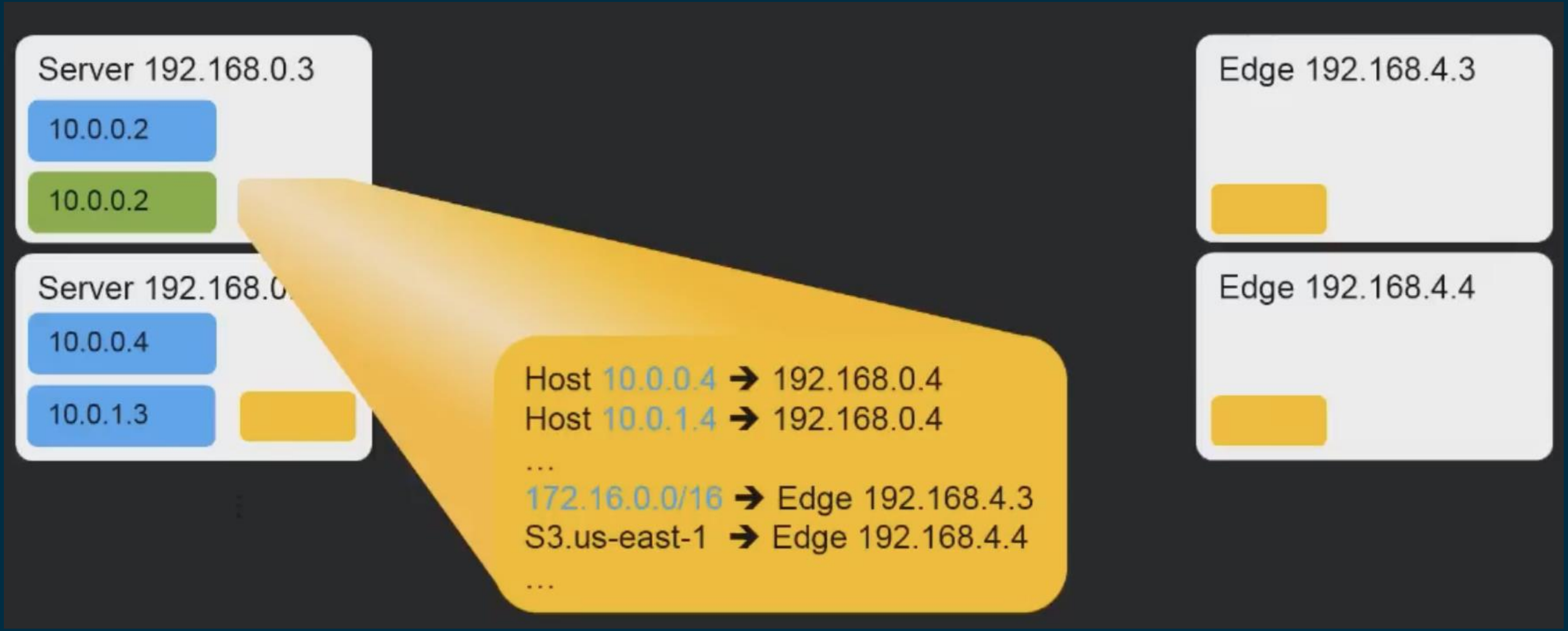
Edges (Blackfoot)



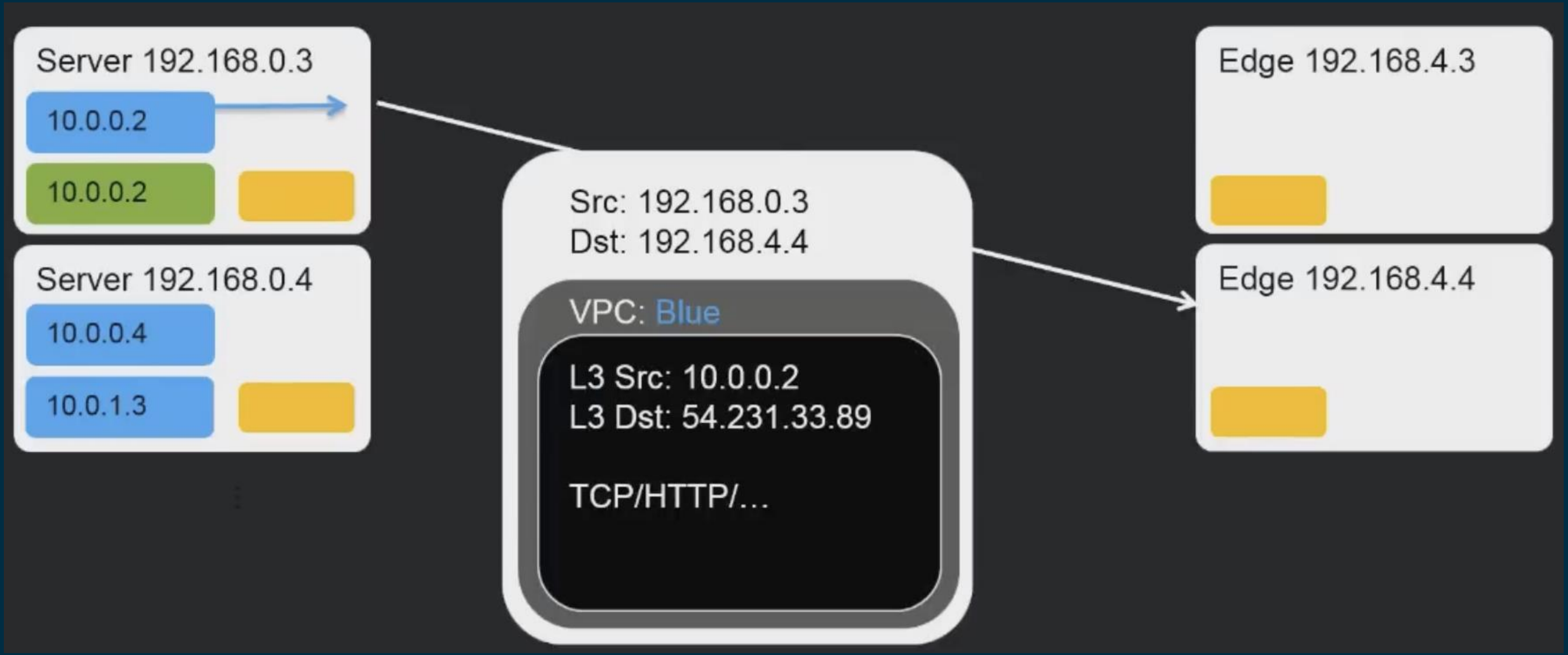
S3 endpoints



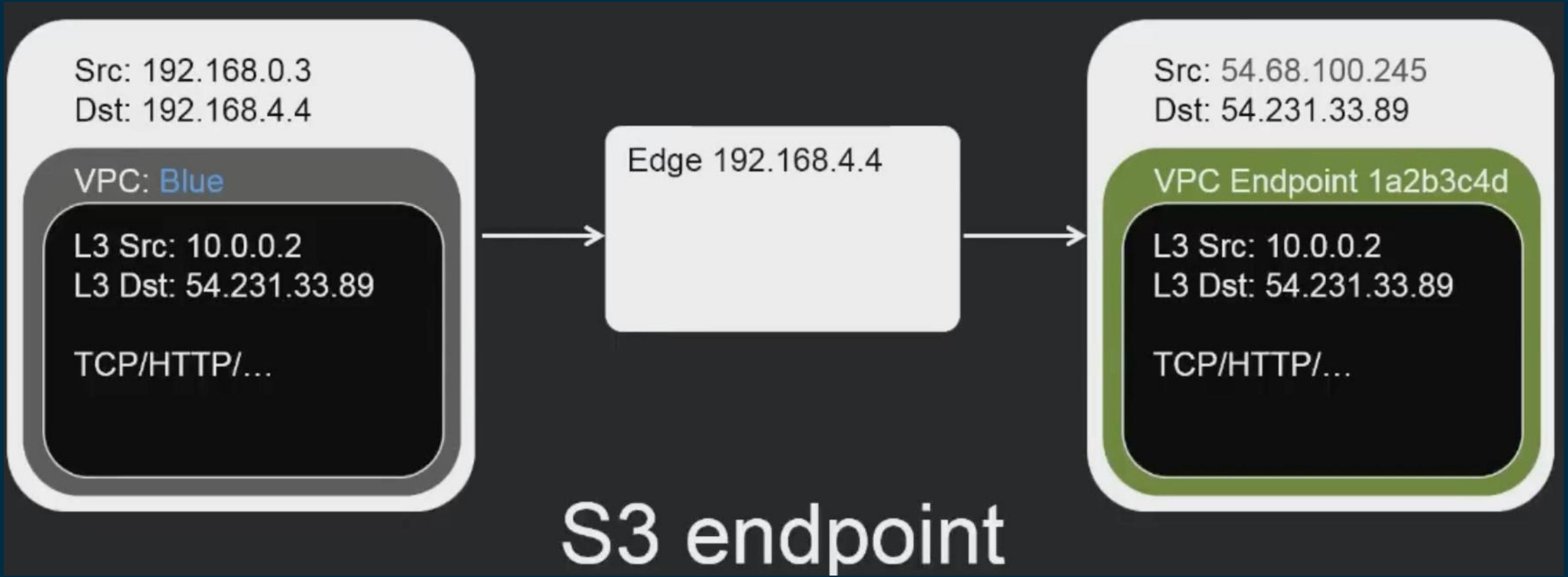
S3 endpoints



S3 endpoints



S3 endpoints



VPC Networking and Flows

- Security Groups include stateful connection tracking
- Flow logs give per-ENI aggregated audit data
- Network Load Balancer can load balance flows natively and transparently in the VPC network
- NAT Gateway brings per-flow stateful NAT to VPC

How flow tracking works

Protocol	Source IP	Destination IP	Source Port	Destination Port
TCP	192.0.2.1	52.84.25.90	33763	443
TCP	192.0.2.1	52.84.25.90	27441	443
UDP	192.0.2.10	205.251.197.26	15732	53
ICMP	192.0.2.1	52.84.25.90	-	-

Protoc ol	Source IP	Destination IP	Source Port	Destination Port	SEQ	ACK
TCP	192.0.2.1	52.84.25.90	33763	443	6532	34224
TCP	192.0.2.1	52.84.25.90	27441	443	18931	45312

First and Next packet sequence and ack. numbers should be relative to each other

How flow tracking works

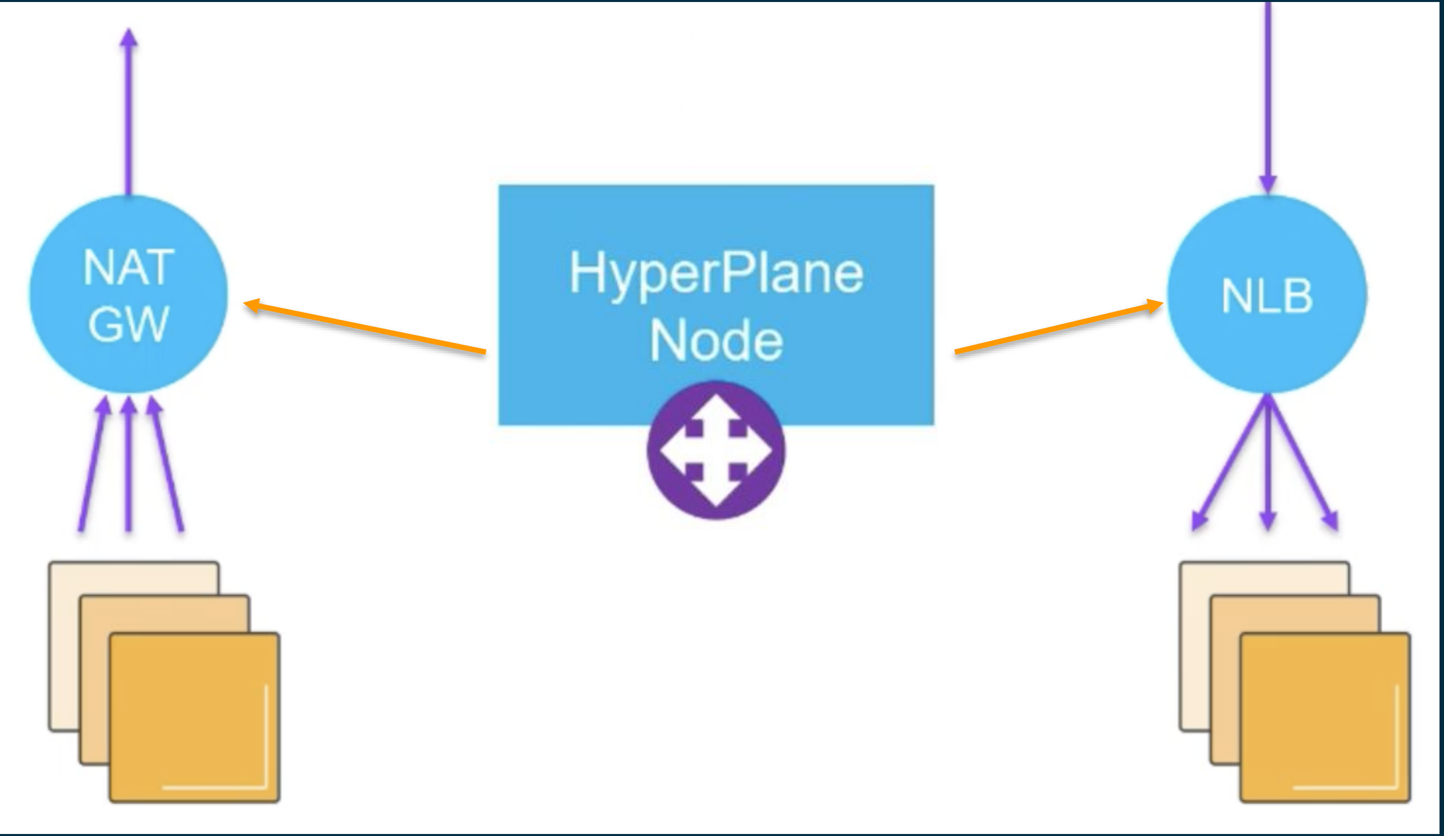
Protocol	Source IP	Destination IP	Source Port	Destination Port	Datagram ID
UDP	192.0.2.10	205.251.197.26	15732	53	5178

Datagram ID should be relative

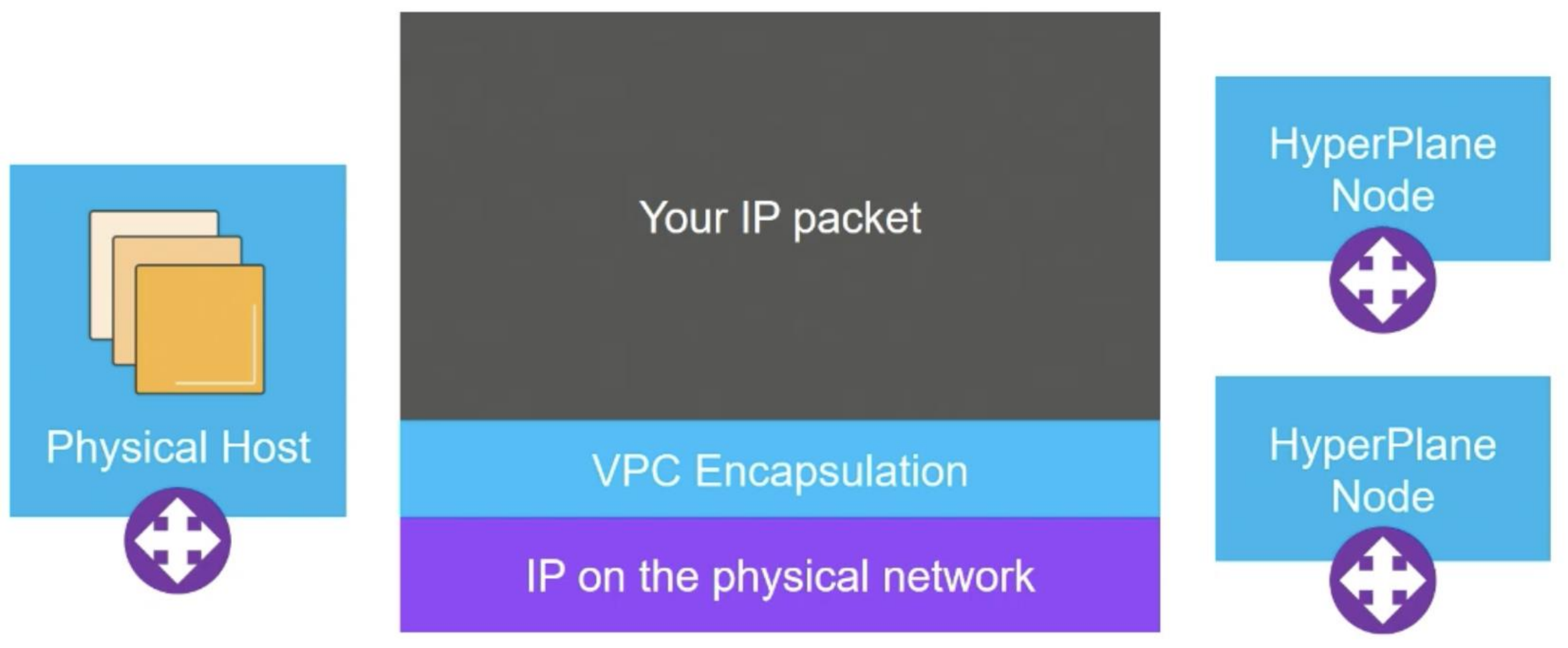
Protocol	Source IP	Destination IP	Bonus embedded header
ICMP	192.0.2.10	205.251.197.26	[Same as previous slides]

ICMP error message should correlate to the previous ICMP packet

NAT Gateway and Network Load Balancer are HyperPlane nodes



HyperPlane



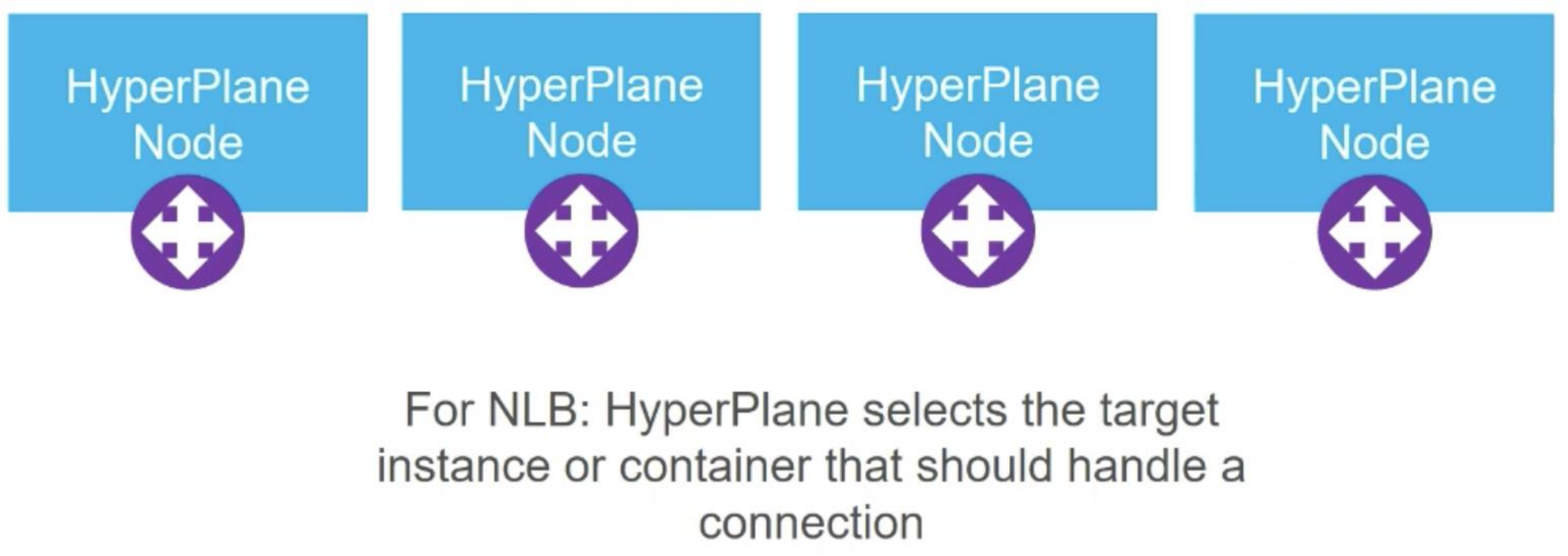
* HyperPlane Node is based on architecture of S3 load balancer

* HyperPlane Nodes are EC2 physical instances   * EFS are HyperPlane Nodes

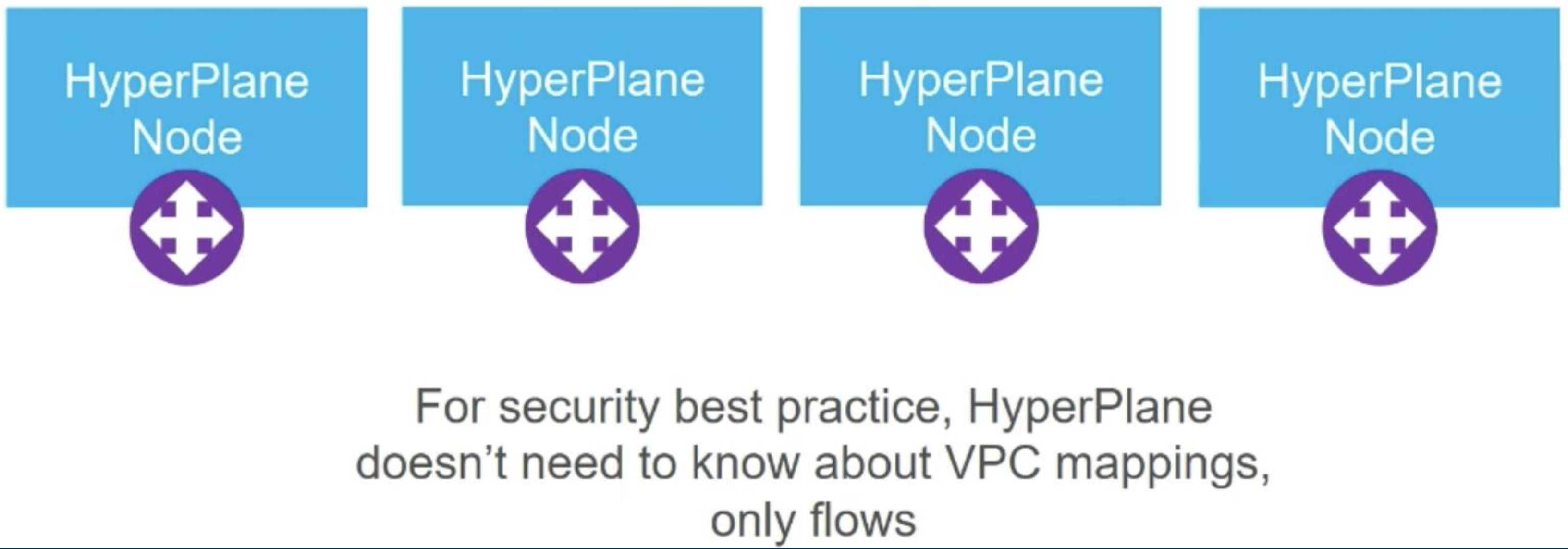
HyperPlane

For NAT: HyperPlane guarantees that connections to the same destination IP / destination port pair have a unique source port

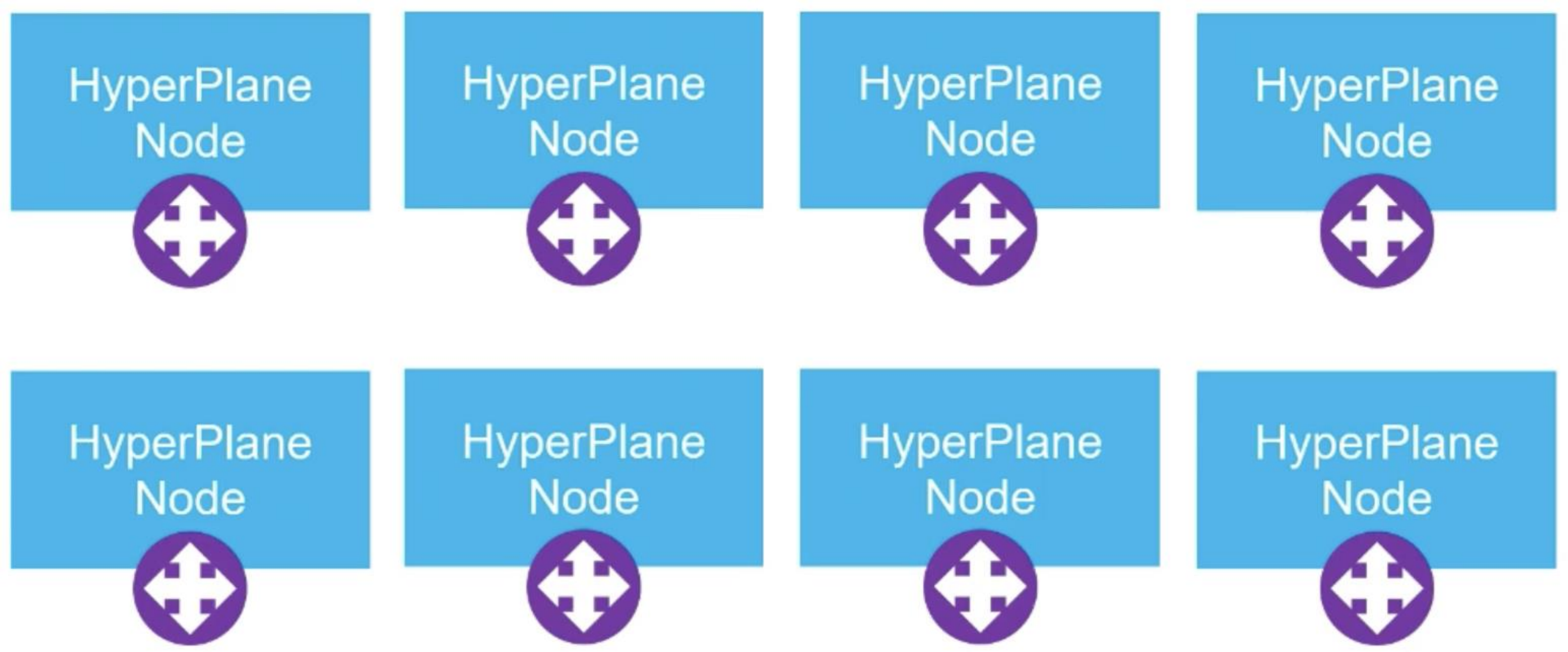
HyperPlane


HyperPlane Node
HyperPlane Node
HyperPlane Node
HyperPlane Node
For NLB: HyperPlane selects the target instance or container that should handle a connection

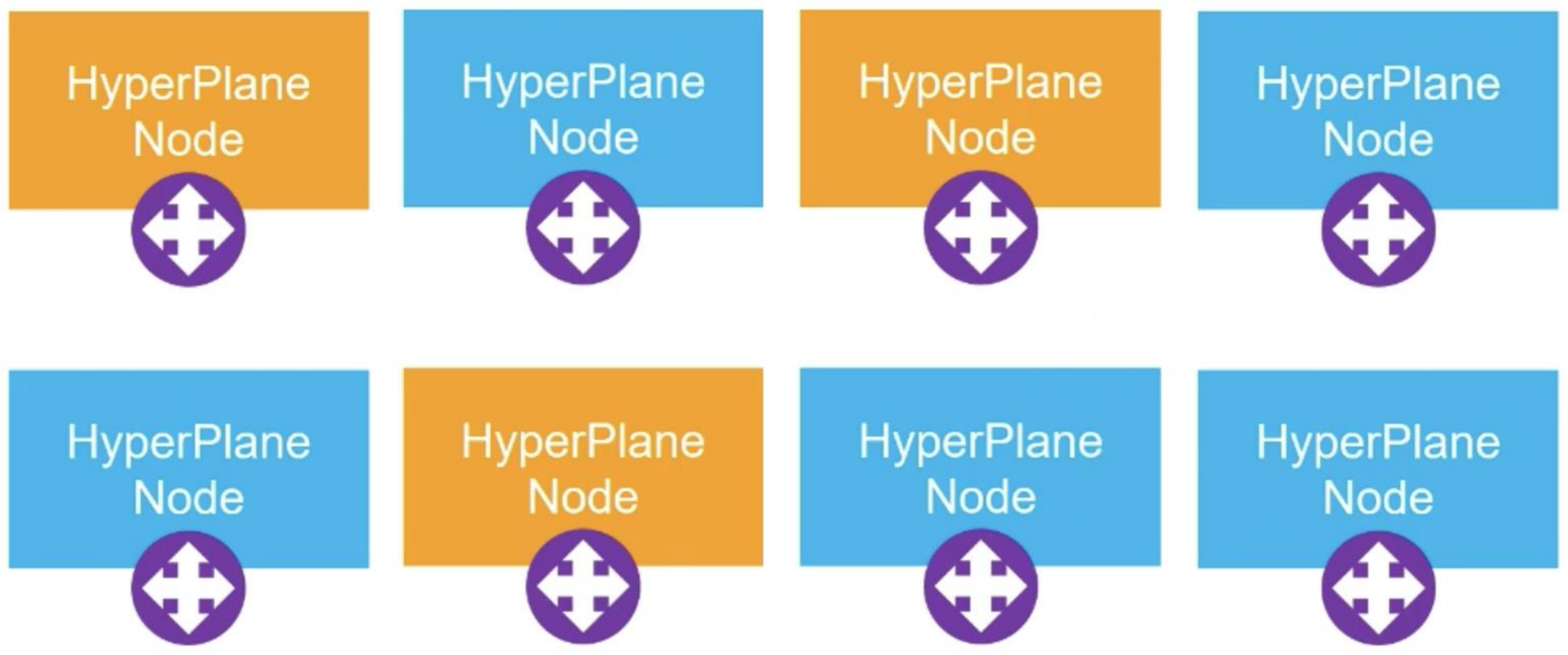
HyperPlane


HyperPlane Node
HyperPlane Node
HyperPlane Node
HyperPlane Node
For security best practice, HyperPlane doesn't need to know about VPC mappings, only flows

Multi-tenancy: HyperPlane and Shuffle Sharding

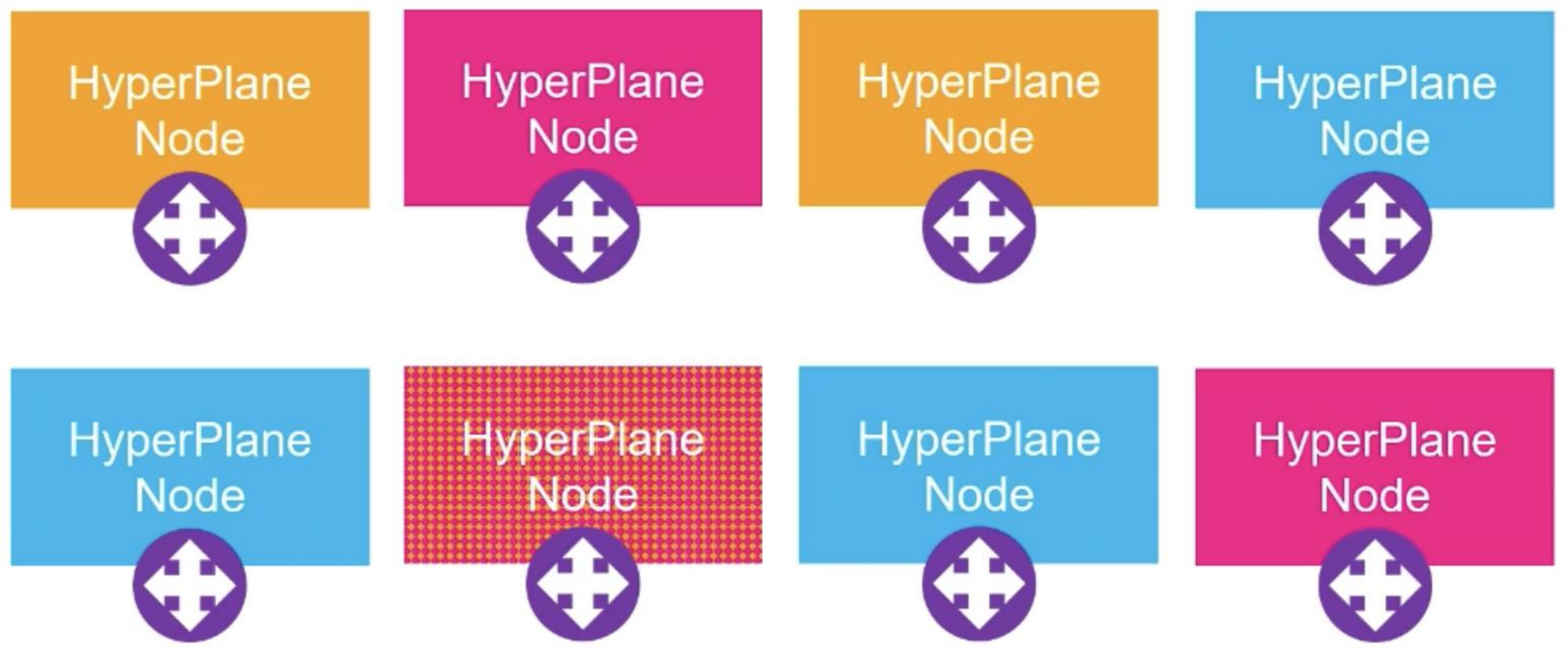


Multi-tenancy: HyperPlane and Shuffle Sharding



3 nodes for Customer A

Multi-tenancy: HyperPlane and Shuffle Sharding



3 nodes for Customer B

Multi-tenancy: HyperPlane and Shuffle Sharding

Potential Overlap	Percentage chance
0	18%
1	54%
2	26%
3	2%

Multi-tenancy: HyperPlane and Shuffle Sharding

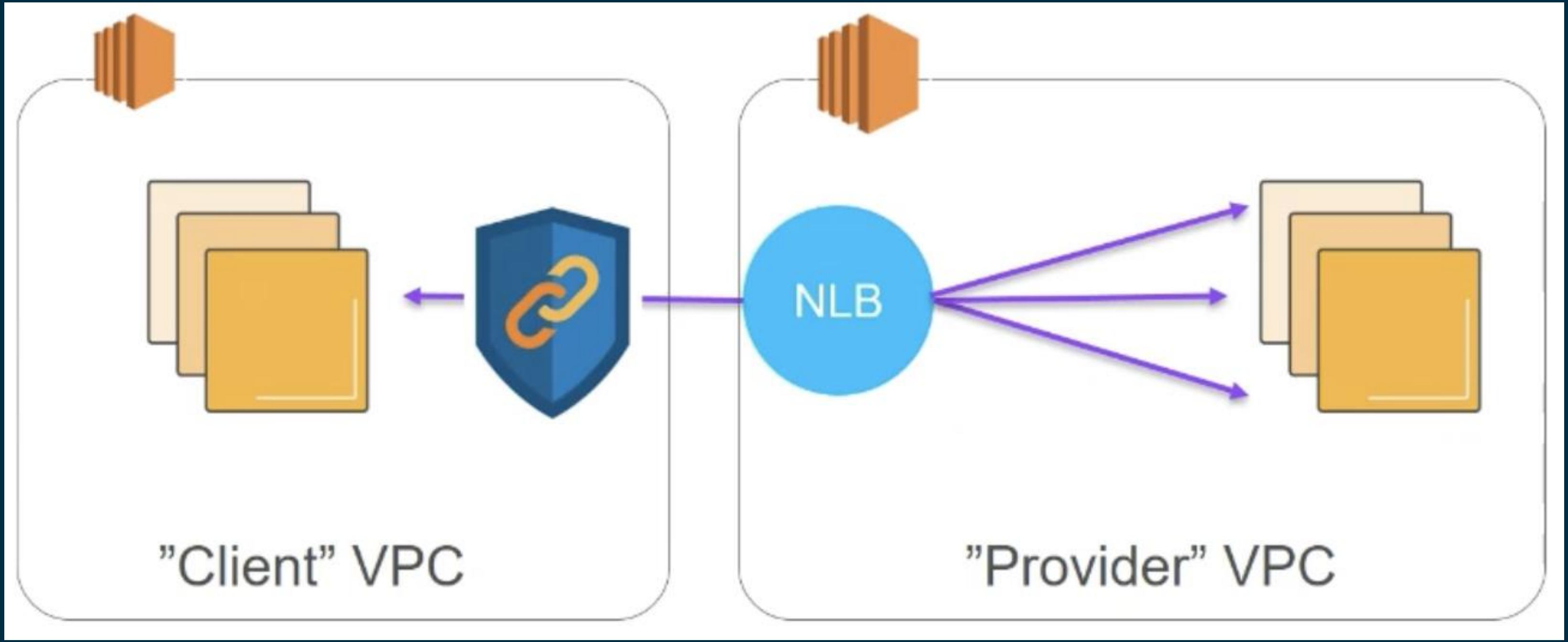
Potential Overlap	Percentage chance
0	77%
1	21%
2	1.8%
3	0.06%
4	0.0006
5	0.00000013

5 nodes for 1 Customer out of 100 nodes

HyperPlane summary

- Based on the S3 Load Balancer
- Used by Elastic Filesystem since launch
- Every HyperPlane resource has 5Gbit/sec of capacity by default, and scales in increments of 5Gbit/sec … to Terabits
- Sub-millisecond latency, hundreds of millions of connections, millions of connections per second

PrivateLink



Thank You!